COMPUTER SCIENCE & IT

DIGITAL LOGIC





Sequential Circuit

DPP 01 Discussion Notes

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Question MCQ

★☆☆

In a S-R flip-flop, the present output is Q = 1. then after applying input S = 0, R = 1 and then S = 0, R = 0, output will be:

$Q(n) = 1$

$S=0, R=1 \longrightarrow Q(n+1) = 0$

$S = 0, R = 0 \longrightarrow Q = 0$

A ✓ Q = 0

B Q = 1

C Q = → invalid state

D None of these

Question **MSQ**

In a JK flip-flop output Q = 0. To change it to Q = 1, the inputs J and K will be:

$Q(n)=0 \longrightarrow 1 \quad Q(n+1)$

A J = 1, K = 1

B J = 1, K = X

C J = 0, K = 0

D J = 0, K = 1

$J=1 \qquad K=X$

$J=1,\ K=1 \rightarrow$
$J=1,\ K=0 \rightarrow$ $\Big\} \rightarrow J=1,\ K=X$

Question **MCQ**

Which of the following is true?

A S = 1, R = 1, is a valid state input for S-R ff

B J = 1, K = 1 is a invalid state input for J-K ff

C Input (0, 0) is hold state input for both S-R as well as JK ff

D None of these

Question **MCQ**

A flip flop has characteristic equation

$Q(n+1) = \bar{A}\bar{Q} + \bar{B}Q$

Then it will be in toggle mode of operation if

$B = 1, A = 0$

$Q(n+1) = \bar{Q} \rightarrow$ toggle mode of operation

$Q(n+1) = \bar{0}\,\bar{Q} + \bar{1}\,Q = \bar{Q} + 0 = \bar{Q}$

A A = 1, B = 1

B A = 0, B = 1 ✓

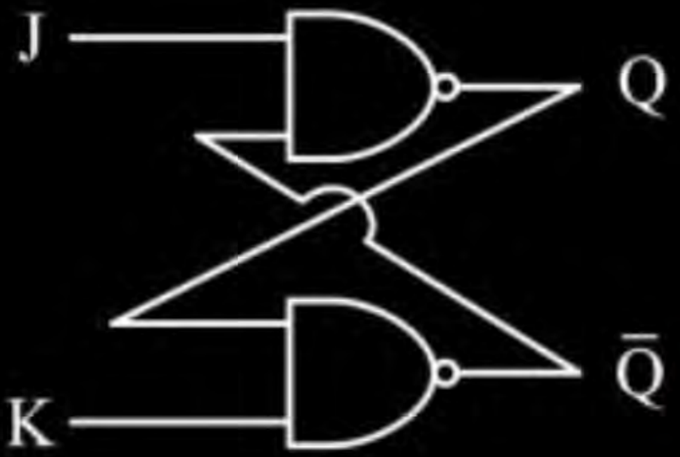
C A = 1, B = 0

D A = 0, B = 0

Question MCQ

★★☆

A logic circuit is as given below:



Then which of the following is true about above circuit?

NAND $S=0, R=0$ invalid state

NOR → $S=1, R=1$ → invalid state

A Race around condition at J = 1, k = 1 ✗

B Race around condition at J = 0, k = 0 ✗

C J = 1, K = 1 is invalid state input ✗

D None of these ✓

Question MCQ



In J-K FF, to change output from 0 to 1 input is changed from

A J = 0, K = 0 to J = 0, K = 1

B J = 0, K = 1 to J = 1, K = 0

C J = 0, K = 1 to J = 0, K = 0

D J = 1, K = 0 to J = 0, K = 1

$J=0, K=1 \longrightarrow 0$

$J=1, K=0 \longrightarrow 1$

$J=0, K=0 \rightarrow 0, 1$

$J=1, K=0 \rightarrow 1$

Question MCQ

Which of the following statement is true?

A Master-Slave ff is used to avoid race-around condition.

B Edge triggered J-K FF has problem of race-around condition when J = 1, K = 1

C Race-around condition occurs in S-R ff with S = 1, R = 1

D None of these

Question MCQ

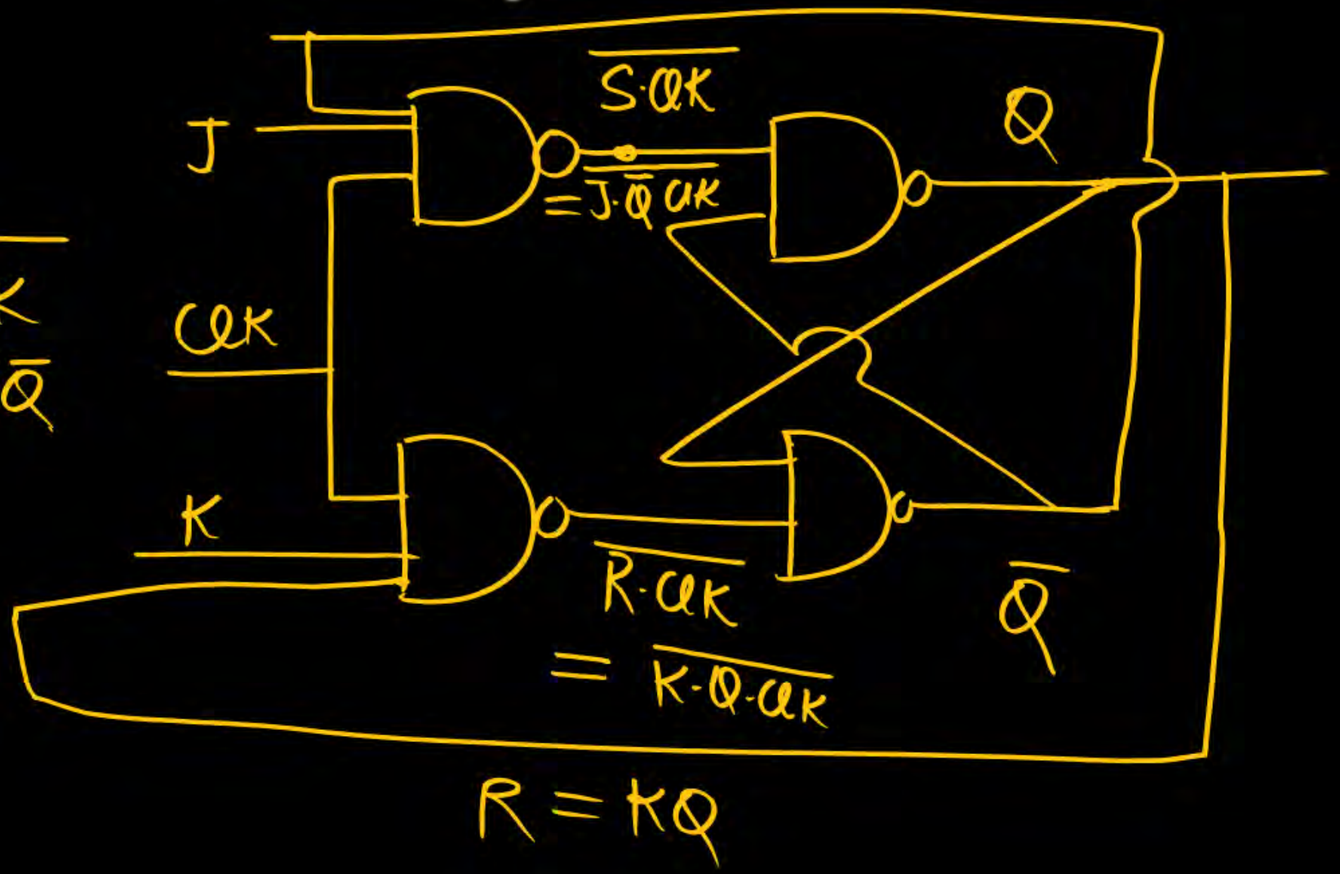
★★☆

To solve the problem of invalid state at input S = 1 and R = 1 in S-R ff, S and R are replaced by

$\overline{S \cdot CLK}$

$\overline{(J\bar{Q}) \cdot CLK}$

$S = J\bar{Q}$



A $S = J\bar{Q}, R = \bar{K}Q$

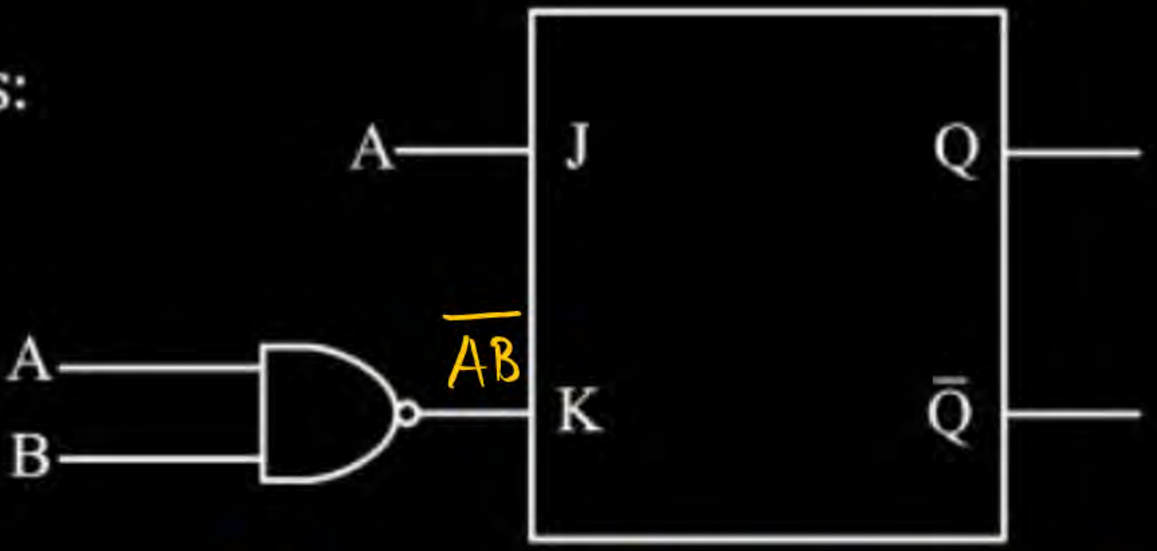
B $S = J\bar{Q}, R = KQ$

C $S = JQ, R = \bar{K}Q$

D $S = JQ, R = K\bar{Q}$

Question **MCQ**

In J-K FF, J and K inputs are given as:



Then which of the following is true?

A $Q(n+1) = A\bar{Q} + AB$

B $Q(n+1) = A\bar{Q} + \bar{B}Q$

C $Q(n+1) = A \oplus B \oplus Q$

D $Q(n+1) = A \oplus B$

$Q(n+1) = J\bar{Q} + \bar{K}Q = A\bar{Q} + \overline{\overline{AB}}Q$

$= A\bar{Q} + ABQ$

$= A[\bar{Q} + (B \cdot Q)]$

$= A[(\bar{Q}+B)\cdot(\bar{Q}+Q)]$

$= A[\bar{Q}+B]$

$Q(n+1) = A\bar{Q} + AB$

Question **MCQ**

★★☆

Which of the following is true?

A $Q(n+1) = S + \bar{R}Q$ is valid for all S and R values → $S \cdot R = 0$

B $Q(n+1) = J\bar{Q} + \bar{K}Q$ is valid when J.K = 1

C $Q(n+1) = S + \bar{R}Q$ is valid when $S \cdot R = 1$

D $Q(n+1) = S + \bar{R}Q$ is valid when $S \cdot R = 0$

Question MCQ

A counter sequence is given as:
2 – 3 – 1 – 0 – 4 – 7, then MOD no. of the counter is

MOD no. = Total no. of different states

MOD no. = 6

A 7

B 6

C 5

D 4

Question **NAT**

A counter sequence is given as:

3 – 2 – 1 – 4 – 6 – 3 – – – – – –

If its starting state is $(100)_2$ then after application of 1049 clock pulses, counter will be at __(__1__)$_{10}$__.

$3-2-1-4-6 \Rightarrow$ MOD no = 5

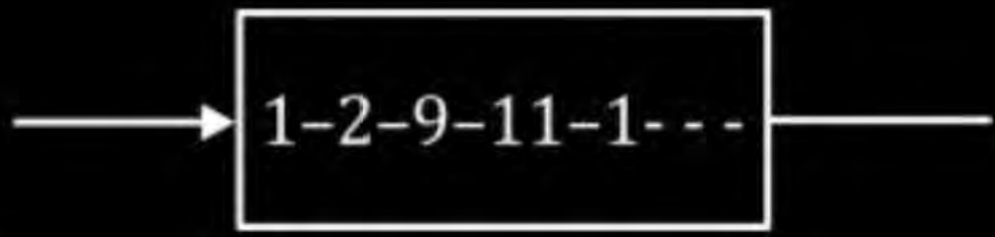
Starting state $= (100)_2 = (4)_{10}$

$(4)_{10} \xrightarrow{5\text{ clk}} (4)_{10} \xrightarrow{5\text{ clk}} (4)_{10}$

$(4)_{10} \xrightarrow[209\times5]{1045} (4)_{10} \xrightarrow{1046} 6 \xrightarrow{1047} 3 \xrightarrow{1048} 2 \xrightarrow{1049} 1$

Question **NAT**

A sequential circuit is as given below:



Value of fout will be 2.75 MHz.

Counter sequence:

1 – 2 – 9 – 11 ⟹ MOD no = 4

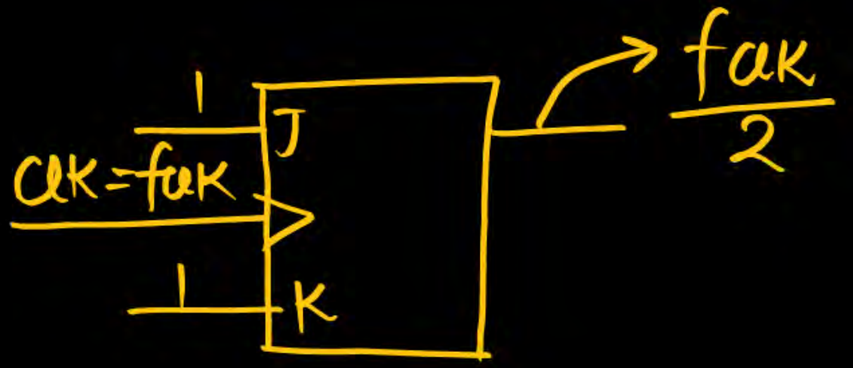
$f_{out} = \frac{11\,MHz}{4}$

$= 2.75\,MHz$

Question MCQ

★★☆

Which of the following is not true?



A In asynchronous counter different FFs are driven by different clock — True

B In asynchronous counter at output of each FF input frequency is divided by 2 — True

C The sequence 0 – 1 – 3 -2 require 2 FFs to design it with asynchronous counter → not true

D Fixed up sequence and fixed down sequence is possible to design with asynchronous counter → True

Question MCQ

★★☆

2 FFs → Total state = $2^2 = 4$

A sequential circuit is as given below:
The above circuit is





$\overline{Q}_0 \rightarrow (1 - 0)$

$Q_0 \rightarrow (0 \rightarrow 1)$

A MOD-4 up counter

B MOD-4 down counter

C MOD-4 neither up nor down counter

D MOD-2 counter

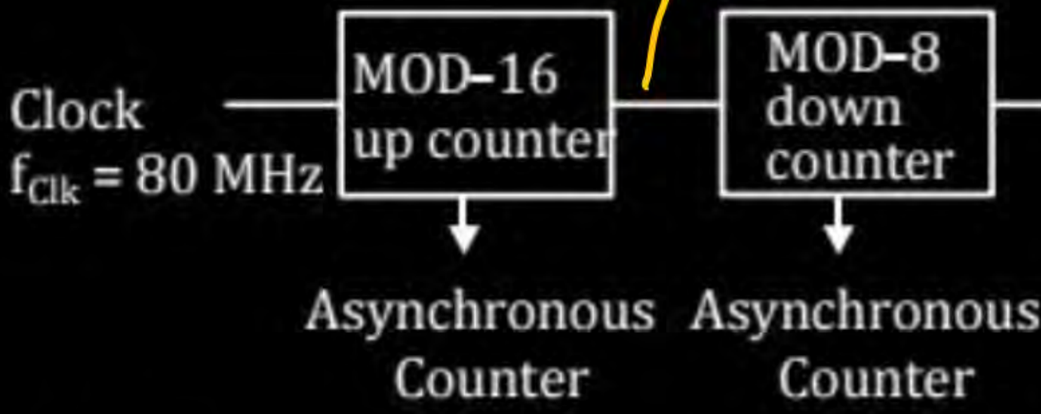
$Q_0 \longrightarrow$ toggles at every clk

$Q_1 \rightarrow$ toggle when $\overline{Q}_0$ $(1-0) \longrightarrow Q_0 \longrightarrow (0-1)$

	Q_1	Q_0
	0	0
clk-1	1	1
clk-2	1	0
clk-3	0	1
clk-4 →	0	0

Question MCQ

A sequential circuit as given below:
Value of f_{out} will be



$f_1 = \frac{f_{clk}}{16} = 5\ MHz$

$f_{out} = \frac{f_1}{8} = \frac{5}{8} = 0.625\ MHz$

A Can not be calculated as one block is up-counter and other is down counter

B 5 MHz

C 0.625 MHz

D 3.33 MHz

Question MCQ

MOD-12 counter can be designed

$M = N_1 \cdot N_2$

A Cascading two MOD-6 counter ✗ [36]

B Cascading MOD-4 counter and MOD-8 counter ✗ [32]

C ✓ Cascading MOD-4 & MOD-3 counter 12

D Cascading MOD-3 & MOD-9 counter [27] ✗

Question NAT

★★☆

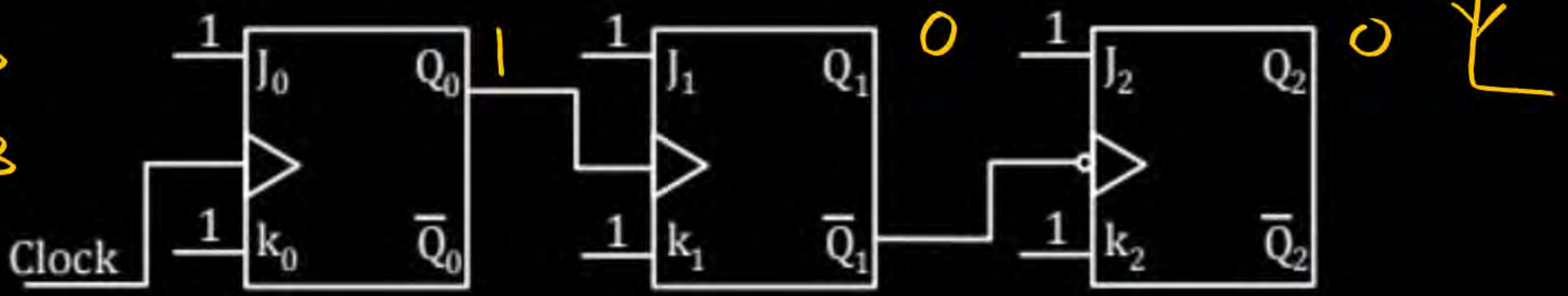
$2^3 = 8 \rightarrow$ MOD-8 down counter

$0 - 7$

A sequential circuit is as given below:

Initially counter is at $Q_0Q_1Q_2 = (100)_2$ then after 2 clock pulses, o/p will be

____(7)$_{10}$____.



$Q_0 \rightarrow$ LSB

$Q_2 \rightarrow$ MSB

	MSB Q_2	Q_1	LSB Q_0
	0	0	1
1st	0	0	0
2nd	1	1	1
3rd	1	1	0

$\rightarrow Q_2Q_1Q_0 = (011) = 3$

100

011

010 ✗

$\overline{Q_1} \rightarrow (1 - 0)$

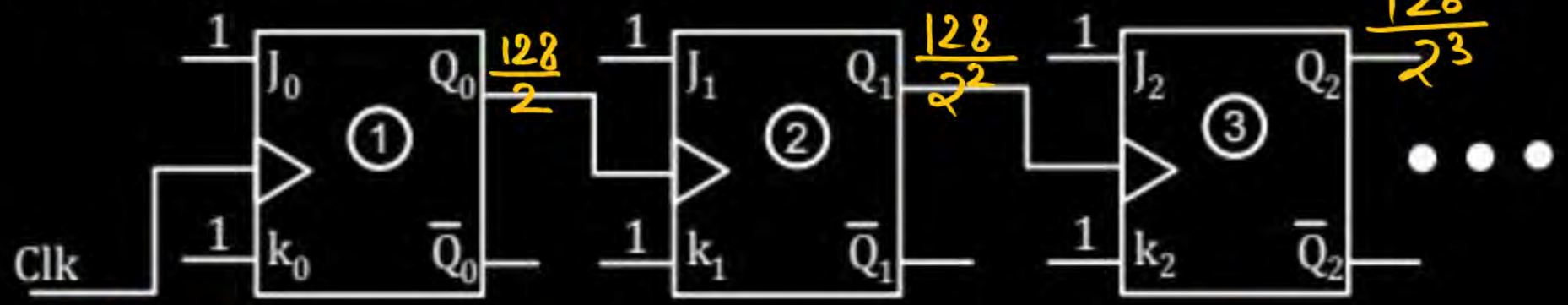
$\downarrow$

$Q_1 \rightarrow (0 - 1)$

Question MCQ

2^{20} = MOD no
↳ down counter

A sequential circuit is as given below:
Total 20 FFs are connected as shown above:
If input clock frequency is 128 MHz then, at output of 16th FF frequency of the waveform will be:



A 8 MHz

B 12.8 MHz

C 1.95 KHz

D 4 MHz

$f_{16} = \frac{128}{2^{16}}$ MHz $= \frac{2^7}{2^{16}}$ MHz $= \frac{1}{2^9}$ MHz

$= \frac{1000}{512}$ KHz $= \frac{1000 \text{ KHz}}{2^9}$

Question NAT

★★☆

Sequential circuit A is designed with 5 FFs and sequential circuit –B is designed with 6-FFs, then maximum MOD no. possible combinedly using sequential circuit-A & sequential circuit-B is 2048.

Circuit-A → 5 FF ⟶ Max^m MOD no. $M_1 = 2^5 = 32$

Circuit-B — 6 FF → Max^m MOD no. $M_2 = 2^6 = 64$

Max^m MOD no combindly $M = M_1 \cdot M_2$ [Cascading]

$= 2^5 \times 2^6$

$= 2^{11} = 2048$

$= 2 \cdot 2^{10}$

Question NAT



To design a MOD-224 counter, minimum no. of FFs required is 8.

$n \geq \log_2 (\text{MOD no.})$

$n \geq \log_2 224 \Rightarrow n \geq 7.80$

$\Rightarrow n \geq 8$

$8\text{FFs} \longrightarrow 2^8$ states = 256 states

Max^m n

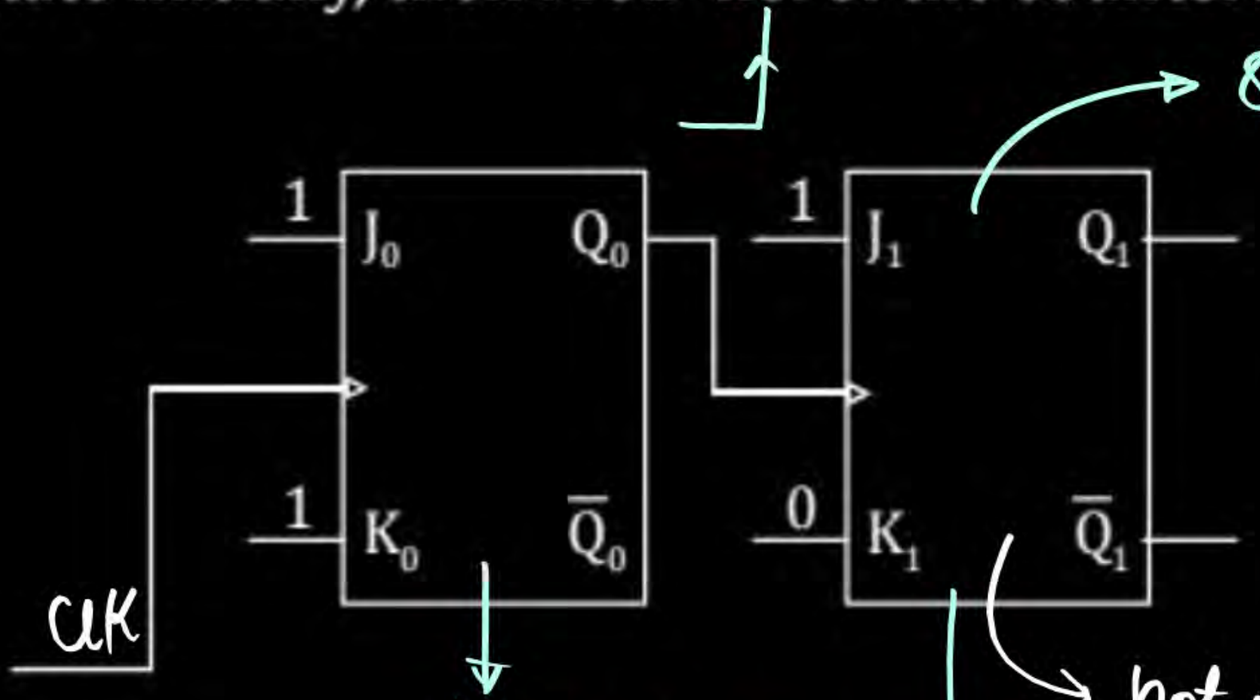
$n\text{FFs} \longrightarrow 2^n \longrightarrow$ 8 state

$7\text{FFs} \longrightarrow 2^7$ states $= (128)$

Question MCQ

A sequential circuit is as given below:

Both the ffs are at reset state initially, then MOD-no. of the counter:



$$Q_1(n+1) = J_1\overline{Q}_1 + \overline{K}_1 Q_1$$
$$= 1\cdot\overline{Q}_1 + \overline{0}\,Q_1$$
$$= \overline{Q}_1 + Q_1$$
$$= 1$$

A MOD-4 counter

B MOD-3 counter

C MOD-2 counter

D None of these

	Q_1	Q_0
	0	0
Clk-1	1	1
Clk-2	1	0
Clk-3	1	1
Clk-4	1	0
Clk-5	1	1
Clk-6	1	0
Clk-7	1	1

0-2-3-0-2-3
-0-2-3



2-3-2-3-2-3

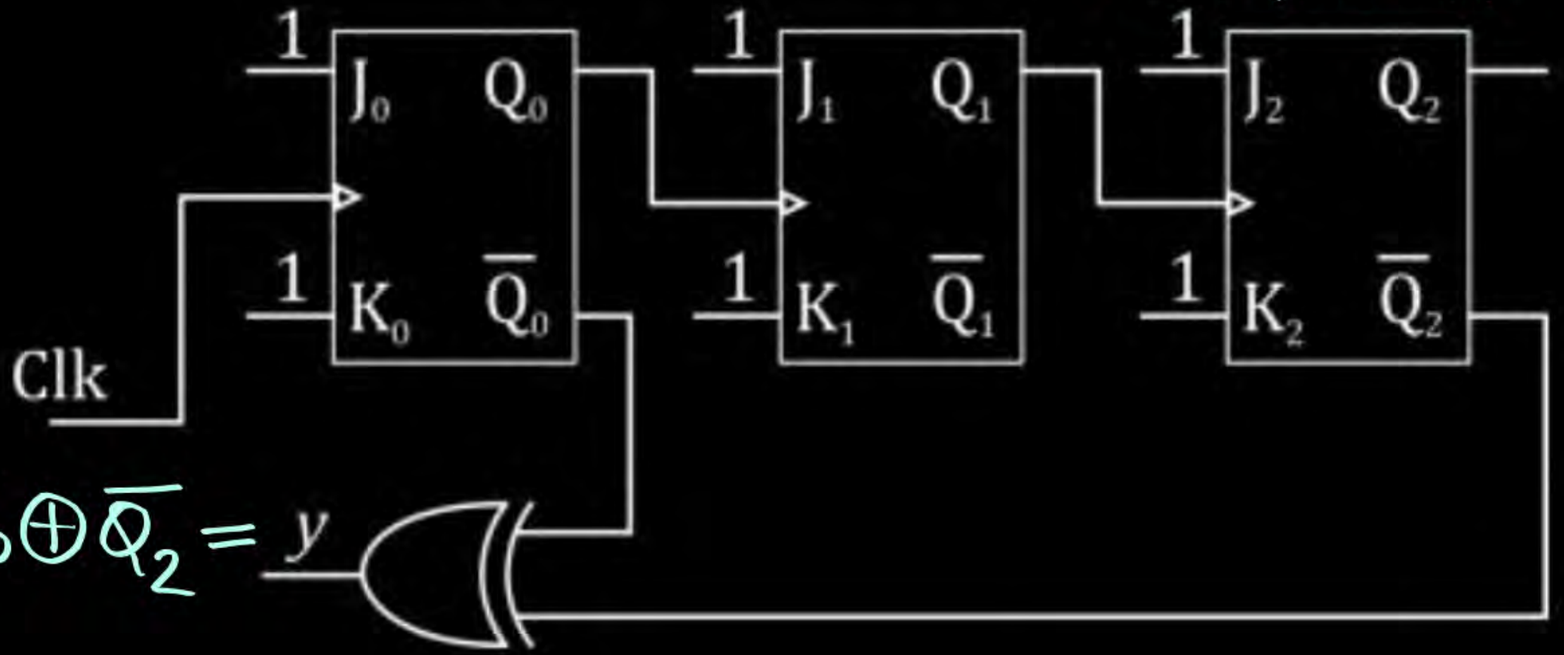
Question NAT

★★☆

A sequential circuit is as given below:

down counter → MOD-2^3

$Q_2 Q_1 Q_0$



We have applied 32 clock pulses then number of times output y toggles is 24

	Q_2	Q_1	Q_0	y
	0	0	0	0
clk-1	1	1	1	0
clk-2	1	1	0	1
clk-3	1	0	1	0
clk-4	1	0	0	1
clk-5	0	1	1	1
clk-6	0	1	0	0
clk-7	0	0	1	1
clk-8	0	0	0	0

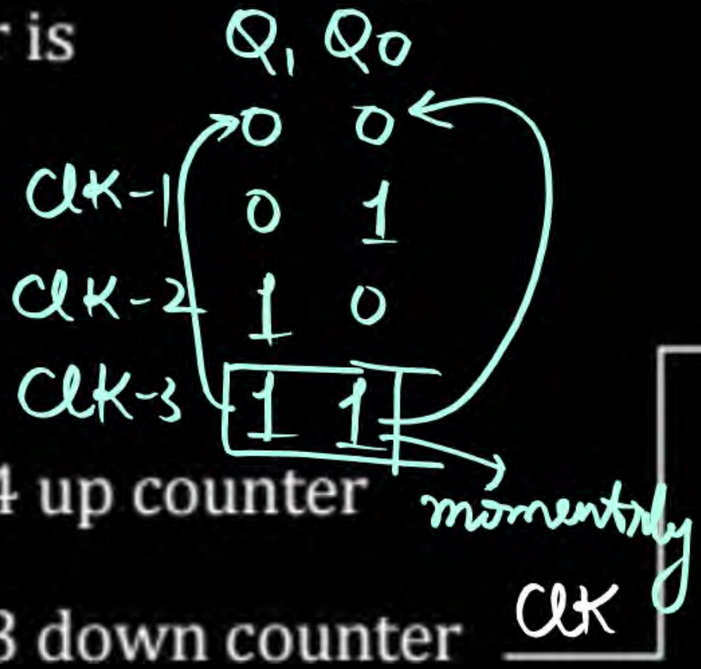
$\longrightarrow 6+6+6+6=24$

Question MCQ

★★☆

A sequential circuit is as given below:

The counter is



A Mod-4 up counter

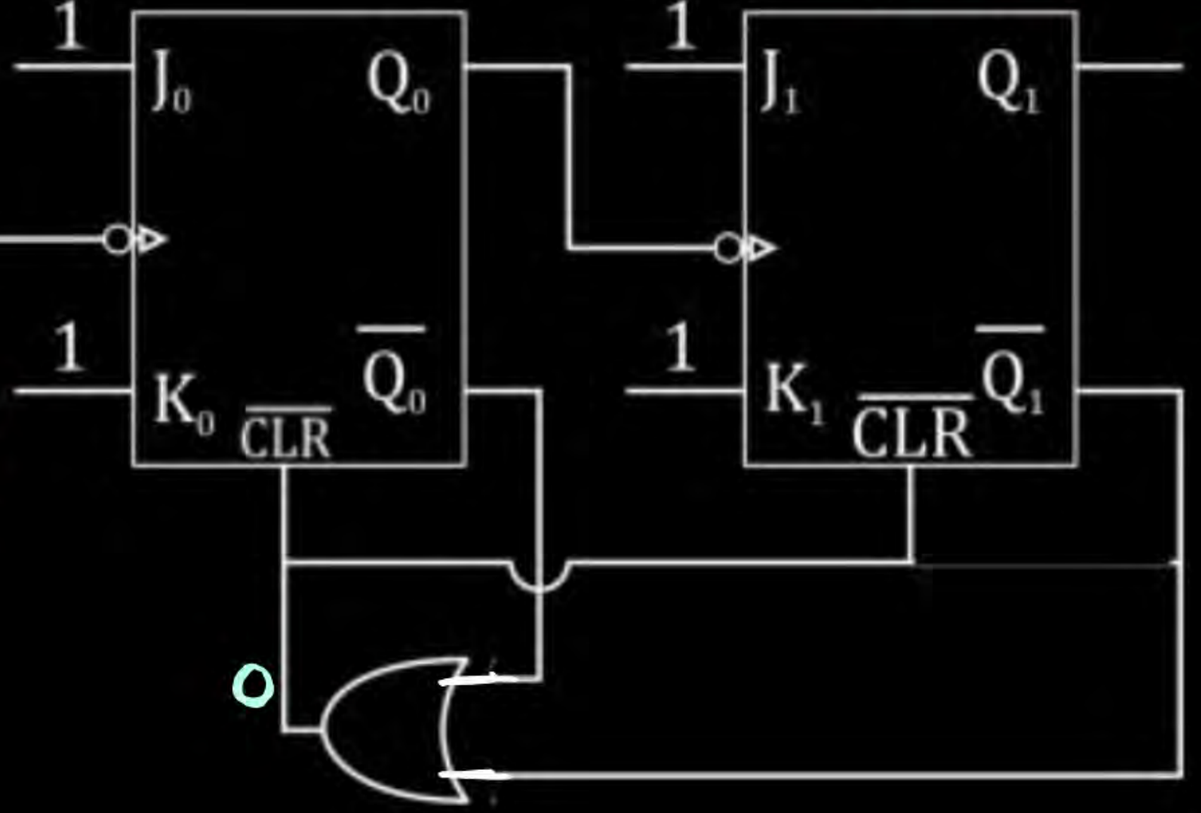
B Mod-3 down counter

C Mod-3 up counter

D Mod-4 down counter

up-counter

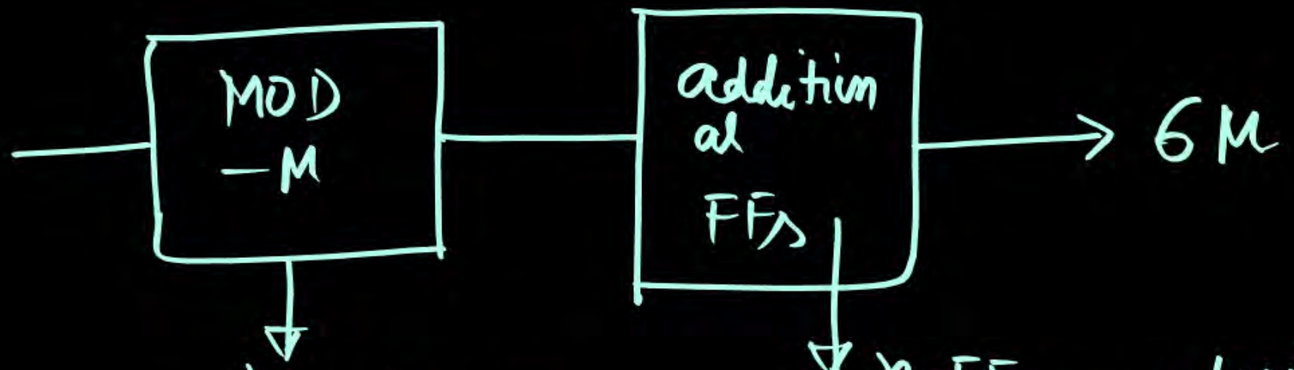
$\overline{Q}_0 = 0$, $\overline{Q}_1 = 0$ → $Q_0 = 1$, CL pin activated, $Q_1 = 1$



Question NAT

Asynchronous counter designed using 4-FFs has Mod number M. By adding FFs in cascade, we changed the MOD number to 6M, then the minimum number of additional FFs used is 3.

Asynchronous Counter → 4FF → Mod no = M



$n \geq 3$

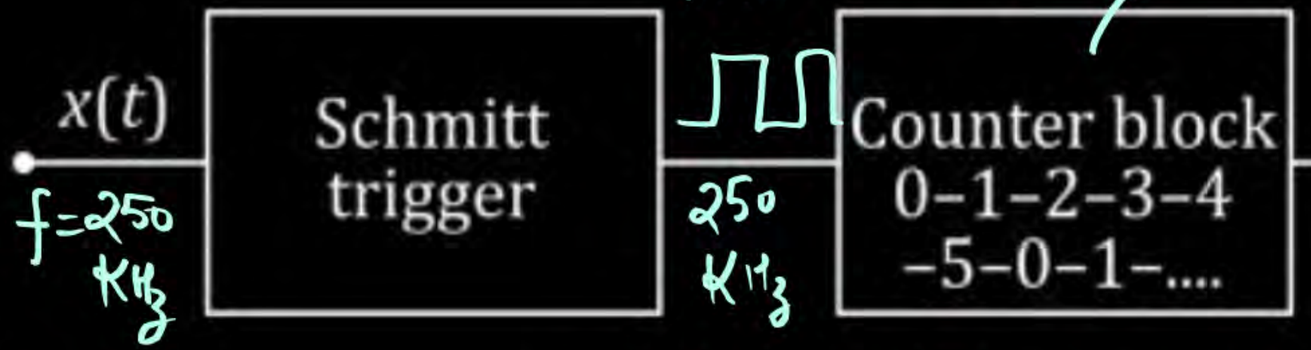
using this we have to design MOD-6 counter and for this $n \geq 3$

Question NAT

★★★

0-1-2-3-4-5 - 0-1-2-3-4-5

A sequential circuit is designed as shown:

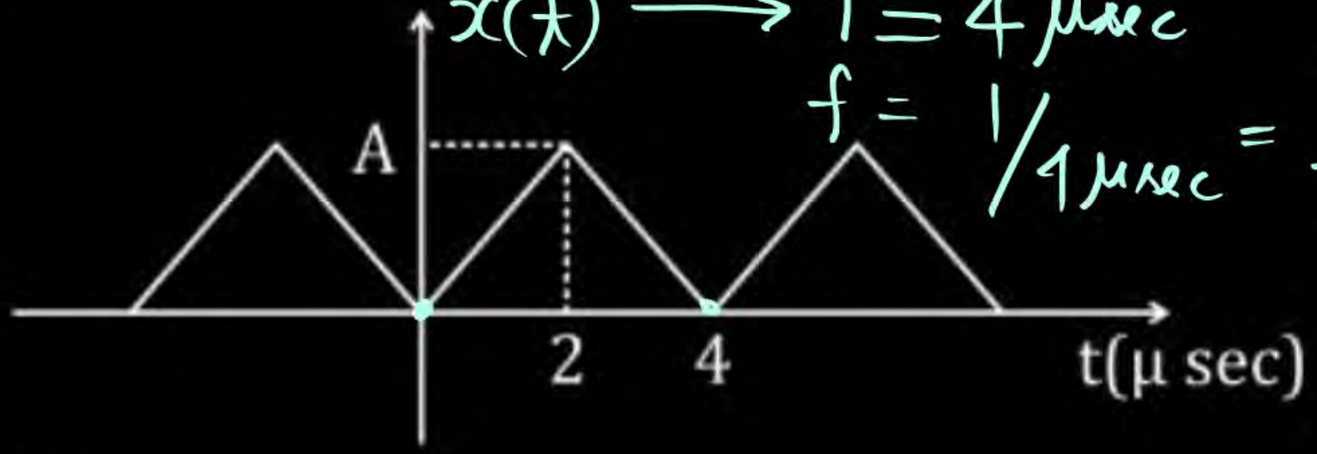


$f_{clk} = 250\ KHz$

MOD-6

$$F_{out} = \frac{f_{clk}}{6} = \frac{250}{6}\ KHz = 41.66\ KHz$$

x(t) is given as



$x(t) \longrightarrow T = 4\ \mu sec$

$$f = \frac{1}{4\ \mu sec} = \frac{10^6}{4}\ Hz = \frac{10^3}{4}\ KHz = 250\ KHz$$

The value of F_{out} 41.67 kHz

Question MCQ

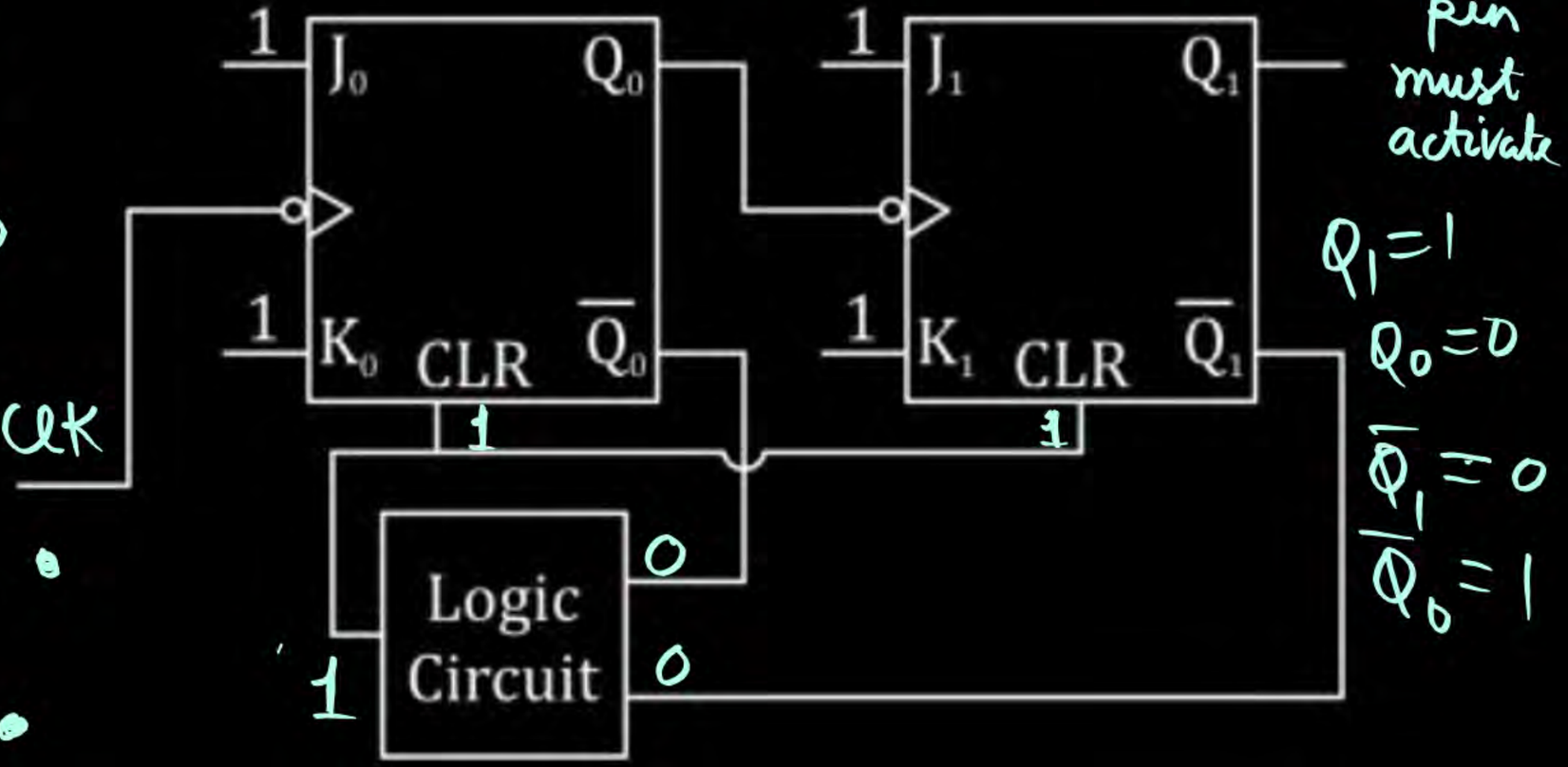
upcounter

A sequential circuit is as given below:

If above sequential circuit has to work as MOD-3 counter then logic circuit will be:

	Q_1	Q_0
	0	0
CLK-1	0	1
CLK-2	1	0
CLK-3	1	1

When $Q_1=1$ & $Q_0=1 \rightarrow$ CLR pin must activate



$Q_1=1$

$Q_0=0$

$\overline{Q}_1=0$

$\overline{Q}_0=1$

A ✗ Two input OR gate

B ✗ Two input AND gate

C ✗ Two input NAND gate

D ✓ Two input NOR gate

Question **MCQ**

Which of the following is true:

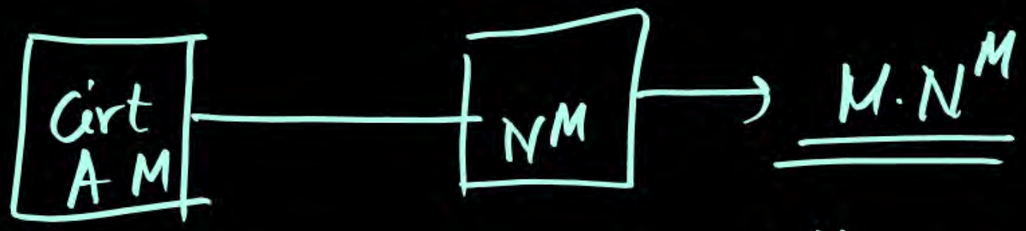
A Asynchronous counters are faster compare to synchronous counters.

B For same MOD-number, synchronous counters require fewer FFs compared to asynchronous counter

C Sequence 0-1-2-3-0 can be designed only in asynchronous counter.

D Sequence 0-1-3-2-0 can be designed only in synchronous counter.

Question MCQ



★☆☆

We have two sequential circuits:

Circuit-A → MOD number M.

Circuit-B → MOD number N.

To design a counter of MOD no. M^N, we can

M^N — counter

$N \cdot N \cdot N \cdots$ M times $= N^M$





$M \cdot M \cdot M \cdot M \cdots$ N times $= M^N$

A Cascade Circuit-A and Circuit-B ✗

B Cascade N Blocks of circuit-A ✓

C Cascade M blocks of circuit-B ✗

D Cascade circuit A with M blocks of circuit-B. ✗

Question MCQ

Lets consider the circuit given below:

Counter is at Q = 0 initially, then after applying clock pulses the sequence generated at $\bar{Q}$ will be

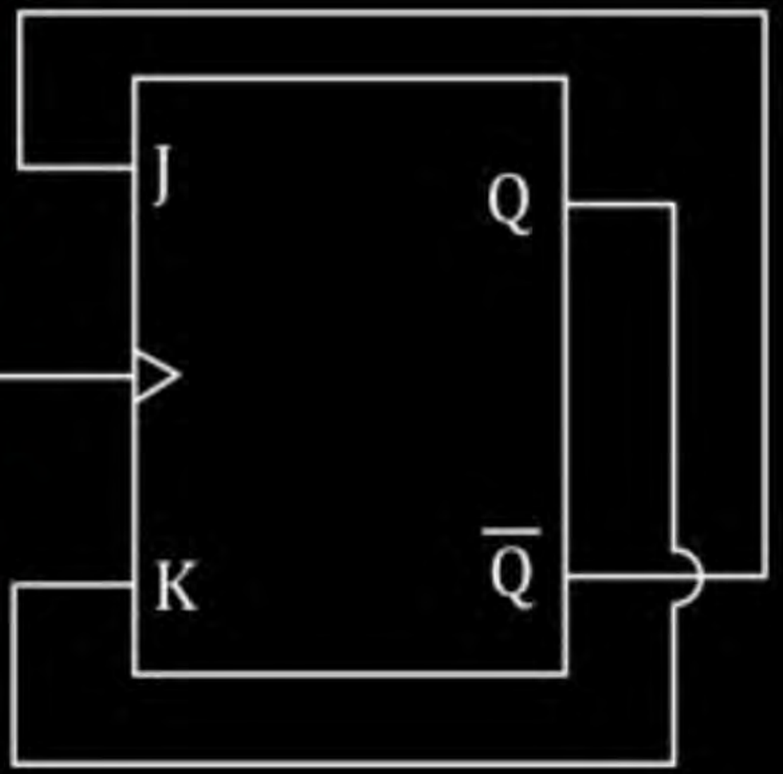
A. 1010101010.....

B. 1100110011005...

C. 0110001010011...

D. 01011011101....

	Q	$\bar{Q}$
1	= 1	0
2	= 0	1
3	= 1	0
4	= 0	1
5	= 1	0
6	= 0	1



$Q(n+1) = J\bar{Q} + \bar{K}Q$

$= \bar{Q}\cdot\bar{Q} + \bar{Q}\cdot Q$

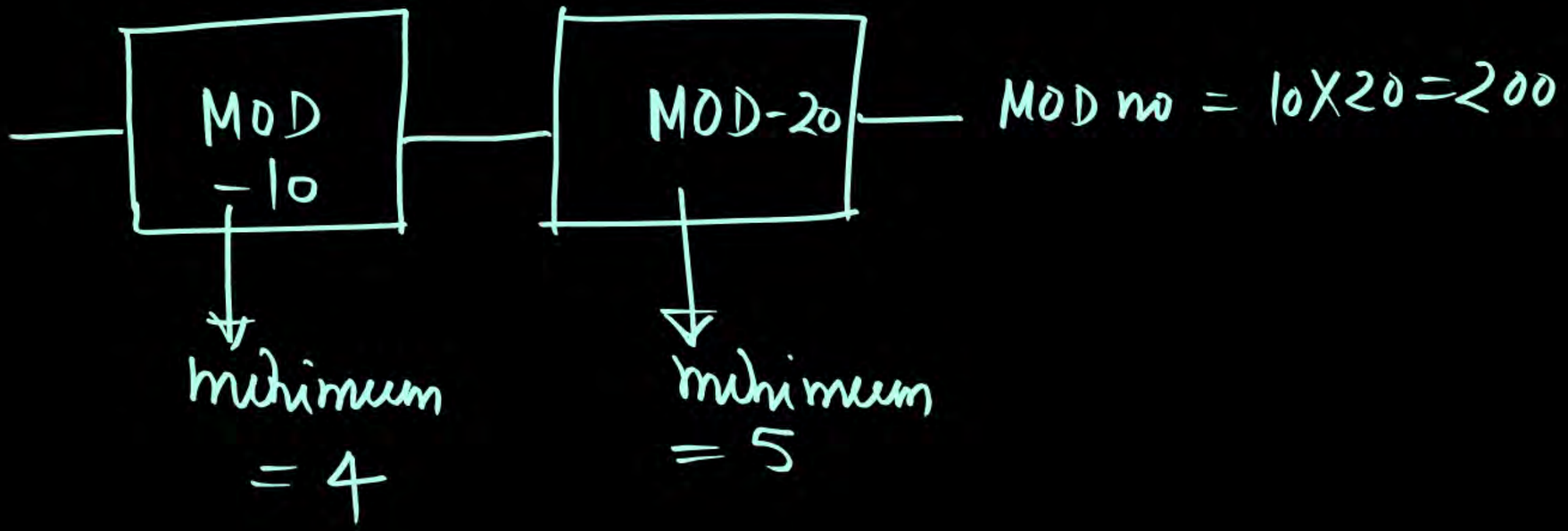
$= \bar{Q} + 0$

$Q(n+1) = \bar{Q}$

Question NAT

★★☆

MOD-10 & MOD-20 counters are cascaded to design MOD-200 counter, then minimum number of ffs required is 9.



Question **MSQ** (A, B)

An asynchronous counter is designed using 10 FFs having MOD No. 1000. Delay of each FF used is 8-n sec, then for which clock frequencies, counter will work properly.

Worst case delay = $10 \times 8 = 80$ nsec

for proper operation



$T_{clk} \geq 80$ nsec

$\frac{1}{T_{clk}} \leq \frac{1}{80\text{nsec}}$

$f_{clk} \leq \frac{10^9}{80}$ Hz

$f_{clk} \leq \frac{10^3}{80}$ MHz

$f_{clk} \leq 12.5$ MHz

- A ✓ 12.5 MHz.
- B ✓ 1.25 MHz.
- C ✗ 125 MHz.
- D ✗ 25 MHz.

Question MCQ

We have M0D-8 asynchronous counter, counting in up-sequence. Delay of each FF is t_d. Initially it is at $(011)_2$ and 1-clock is applied then decoding errors that will appear →

MOD-8 (0-7)

011

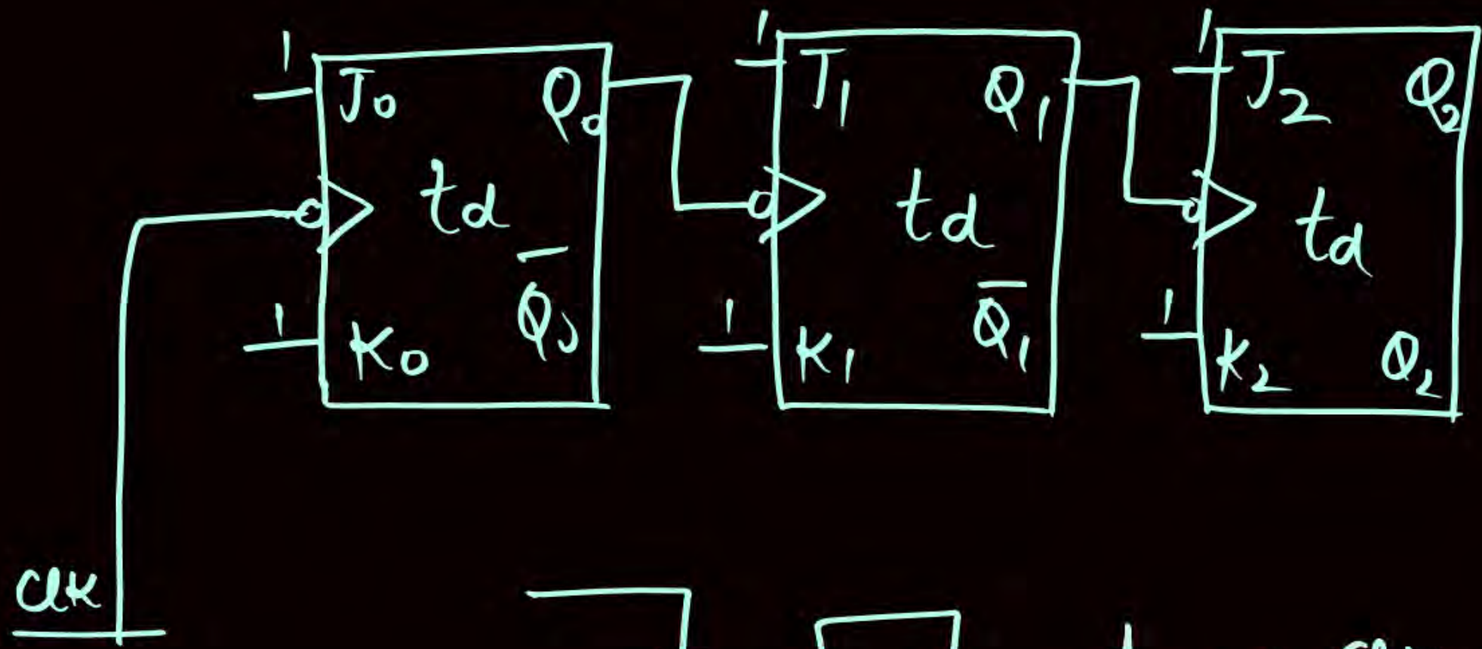
100

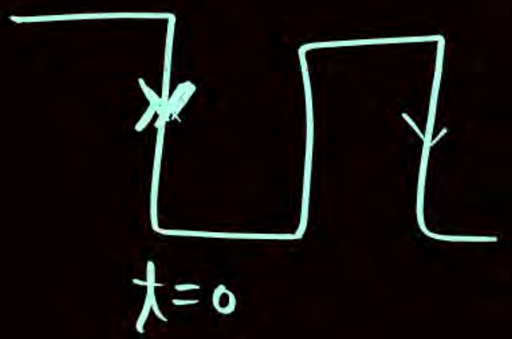
A $(010)_2$ & $(000)_2$

B $(000)_2$ & $(101)_2$

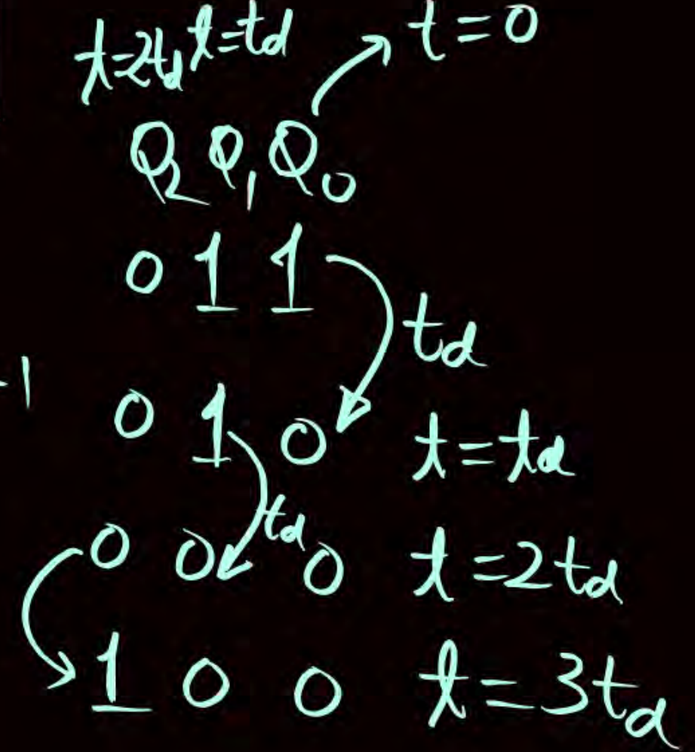
C $(101)_2$ & $(010)_2$

D $(010)_2$ & $(110)_2$





$t = 0 \rightarrow clk-1$



Question NAT

We have a MOD-128 down counter starting with state $(111)_{10}$. After application of 370 clock pulses, counter will be at $_(\underline{125})_{10}_$

Starting state $(111)_{10} \xrightarrow{128 \text{ clk}} (111)_{10} \xrightarrow{128 \text{ clk}} (111)_{10} \xrightarrow{111 \text{ clk}} (0)_{10}$

$370 - 256 = (114)$

MOD-128 down counter → 0 — 127

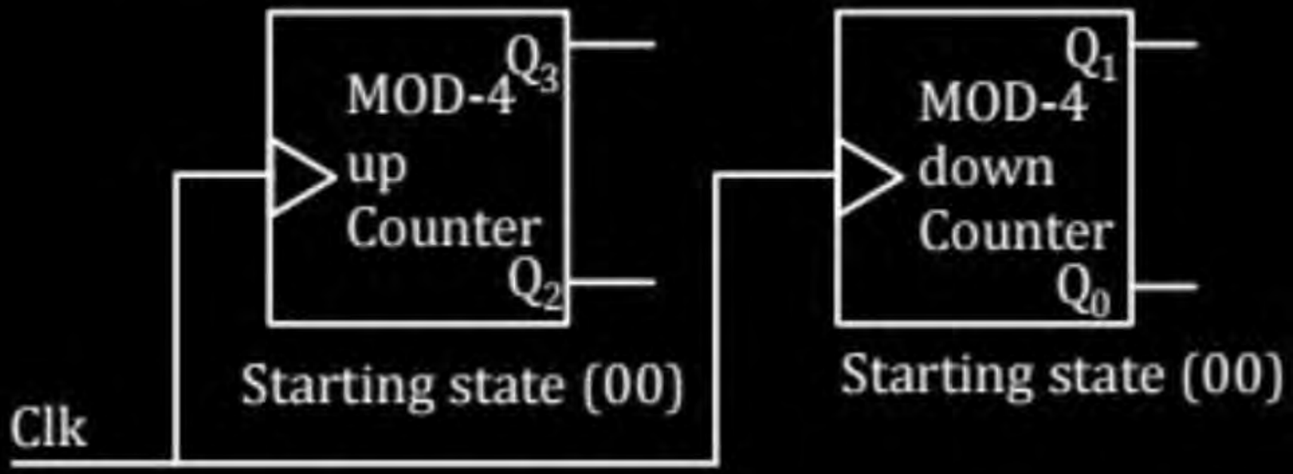
$(125)_{10} \xleftarrow{3} 126 \xleftarrow{2} 127 \xleftarrow{1} (0)_{10}$

127 — 126 — 125 . . . . — 0 — 127 — 126 —

Question MCQ

★★☆

A sequential circuit is as given below:
After applying clock pulses, which count sequence will not appear at output $Q_3Q_2Q_1Q_0$ is



A $(0)_{10}$

B $(7)_{10}$

C $(10)_{10}$

D $(11)_{10}$

	Q_3	Q_2	Q_1	Q_0	
	0	0	0	0	$\rightarrow (0)_{10}$
CLK-1	0	1	1	1	$\rightarrow (7)_{10}$
CLK-2	1	0	1	0	$\rightarrow (10)_{10}$
CLK-3	1	1	0	1	$\rightarrow (13)_{10}$
CLK-4	0	0	0	0	

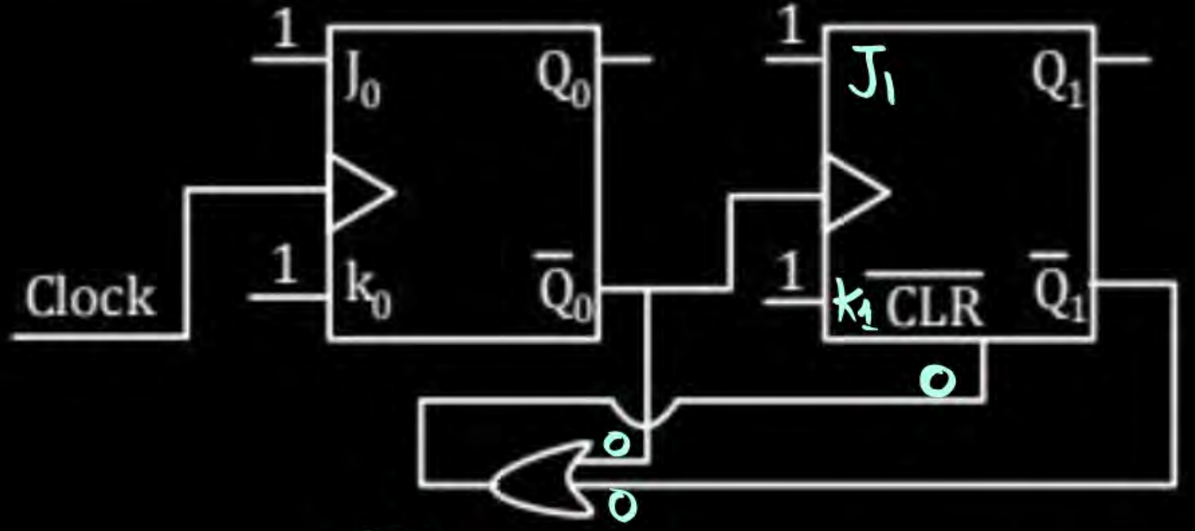
0-3-2-1-0-

Question MCQ

★★☆

Asynchronous up counter

We have sequential circuit as given below: -
The above circuit is



A MOD-3 counter

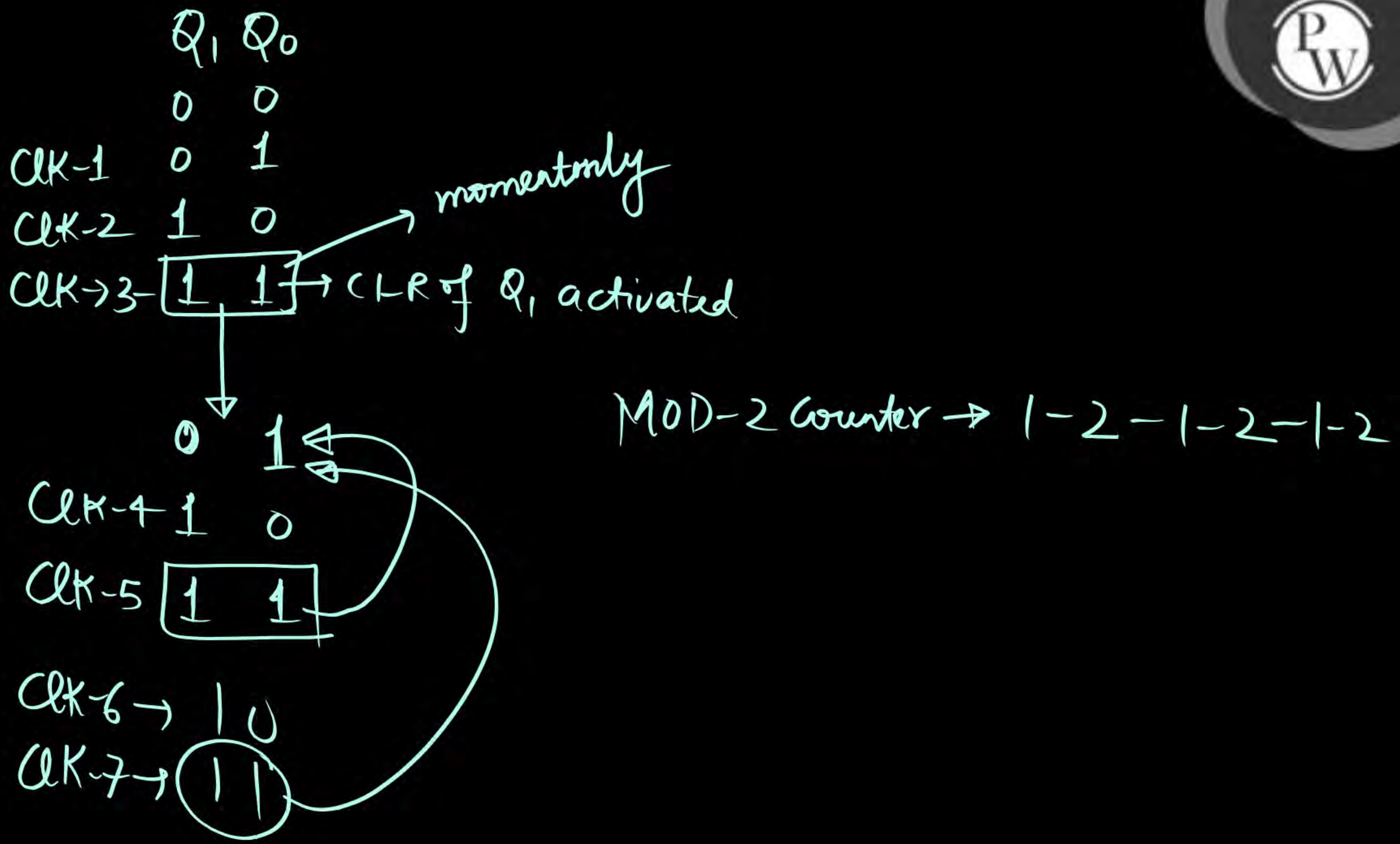
B MOD-2 counter

C MOD-4 counter

D None of these

$\overline{Q}_0 = 0$, $\overline{Q}_1 = 0$ → CLR of Q_1 will activated.

$Q_0 = 1$, $Q_1 = 1$ → CLR pin of Q_1 will activated.


Q_1 Q_0
0 0
Clk-1 0 1
Clk-2 1 0
Clk→3- 1 1
momentarily
CLR of Q_1 activated
0 1
Clk-4 1 0
Clk-5 1 1
Clk-6→ 1 0
Clk-7→ 1 1
MOD-2 Counter → 1-2-1-2-1-2

Question NAT

MOD-32 up counter has starting state $(00110)_2$ and MOD-16 down counter has starting state $(0110)_2$. After application of 75 clock pulses up counter is in state M and down counter is in state N then value of M + N is __$(28)_{10}$__.

0-31 ← MOD-32 - up counter → starting state $(00110)_2 = (6)_{10}$

MOD-16 → down counter → starting state → $(0110)_2 = (6)_{10}$

↳ 0-15 ⟹ 15-14-13-12 ··· 0→15

up $(6)_{10} \xrightarrow{32\text{ clk}} (6)_{10} \xrightarrow{32\text{ clk}} (6)_{10} \xrightarrow{11\text{ clk}} (17)_{10} = M$

down $(6)_{10} \xrightarrow{16\text{ clk}} (6)_{10} \xrightarrow{16\text{ clk}} (6)_{10} \xrightarrow{16} (6)_{10} \xrightarrow{16\text{ clk}} (6)_{10} \xrightarrow{6\text{ clk}} (0)_{10} \xrightarrow{1\text{ clk}} (15)_{10}$

$N = (11)_{10} \xleftarrow{4\text{ clk}}$

Question MCQ

Which of the following is true about decoding error in asynchronous counter?

A It is removed by using additional signalling

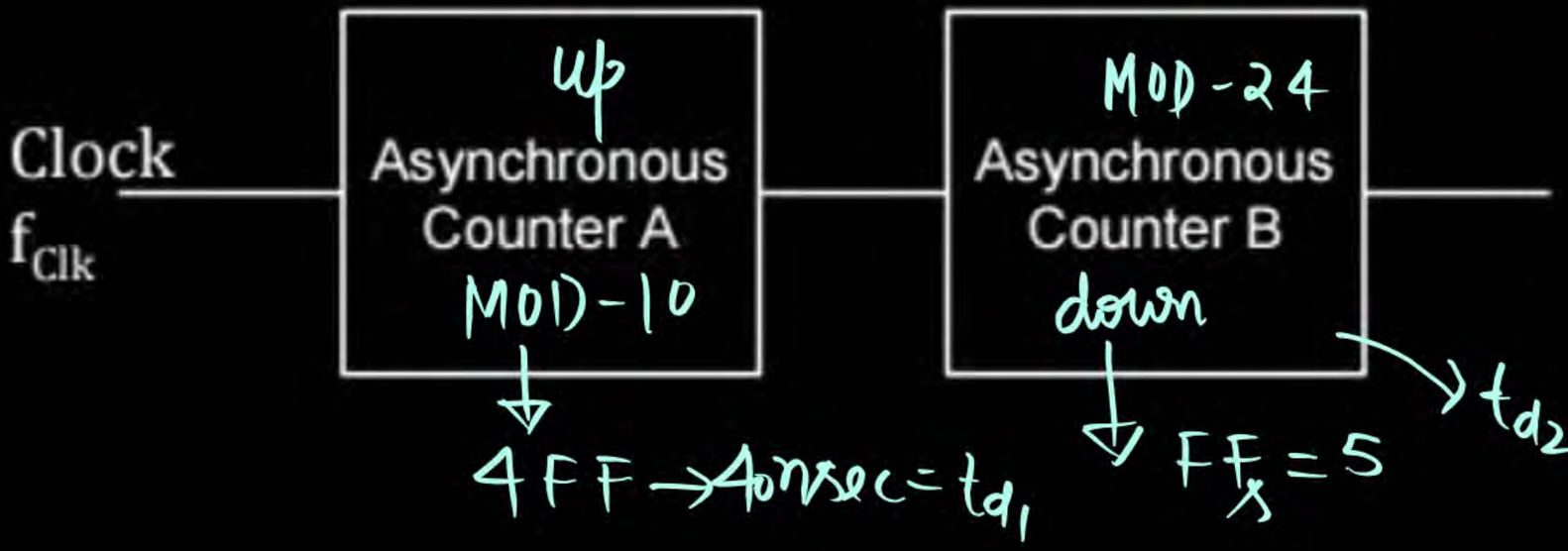
B It can be avoided using strobe signal

C Decoding error is present in asynchronous as well as in synchronous counter

D None of these

Question **NAT**

A sequential circuit is as given below:



$T_d = t_{d_1} + t_{d_2} = 90nsec$

$T_{clk} \geq T_d \quad f_{clk} \leq \frac{1}{T_d}$

$f_{clk} \leq \frac{1}{90nsec}$

$f_{clk} \leq \frac{10^9}{90}$ Hz

4 FF → $40nsec = t_{d_1}$

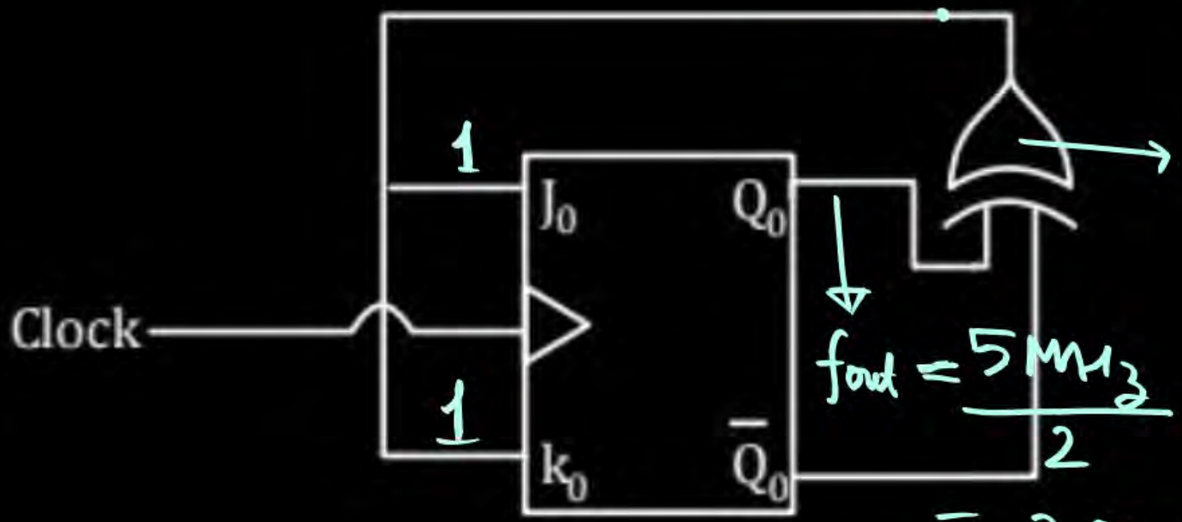
$FF_s = 5$

$f_{clk} \leq \frac{100}{9}$ MHz

Counter A is MOD-10 up counter and counter B is MOD-24 down counter. FF used in designing has equal delay of t_d = 10nsec. Then maximum clock frequency value of f_{Clk} that can be used to have proper operation of the circuit is 11.11 MHz.

Question **NAT**

A sequential circuit is as given below:
If input clock frequency is 5 MHz, the frequency of output waveform at output Q_0 is 2.5 MHz.



$1 = Q_0 \oplus \overline{Q}_0$

$Q_0(n+1) = J_0\overline{Q}_0 + \overline{K}_0 Q_0$

$Q_0(n+1) = 1\cdot\overline{Q}_0 + \overline{1}\, Q_0 = \overline{Q}_0$

Q_0

0

$f_{out} = \frac{5\text{ MHz}}{2}$

$= 2.5$ MHz

CLK-1 1

CLK-2 0

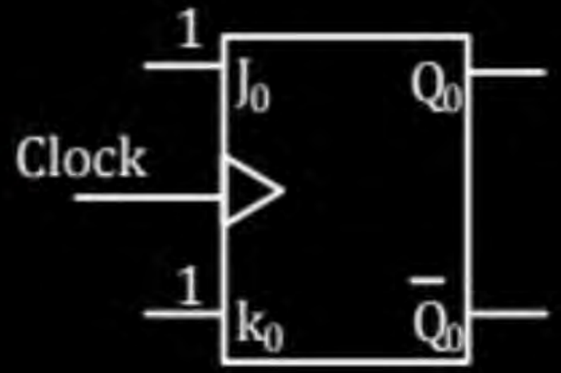
CLK-3 1

→ MOD-2 counter

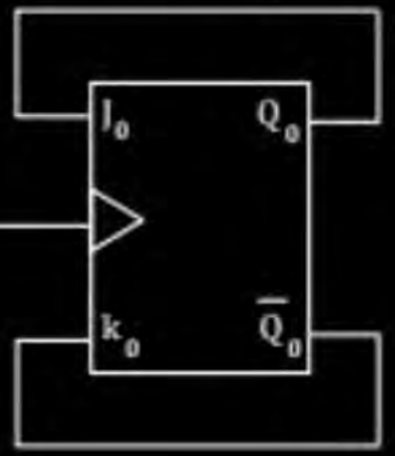
Question **MSQ** (A, C, D)

Which of the following circuit represents toggle mode of operation.

A

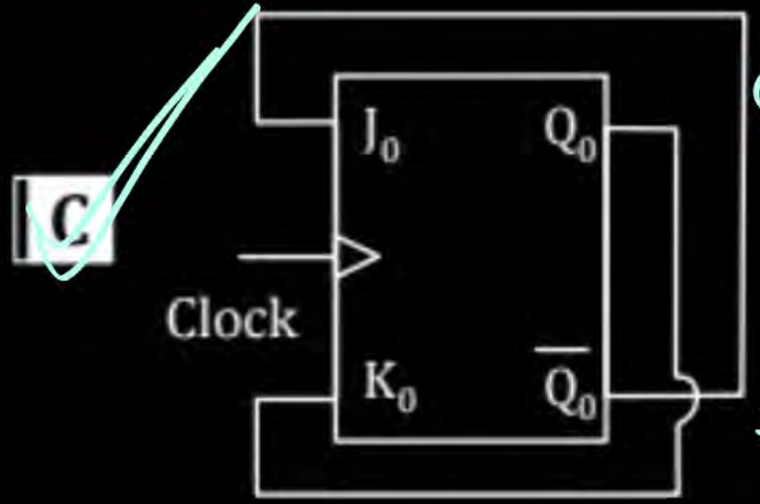


B



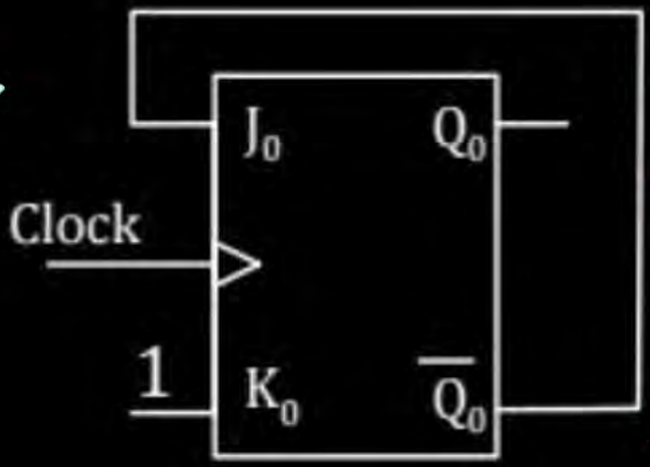
$$Q_0(n+1) = J_0 \cdot \overline{Q_0} + \overline{K_0} Q_0$$
$$= Q_0 \cdot \overline{Q_0} + Q_0 \cdot Q_0$$
$$= 0 + Q_0 = Q_0$$

C



$$Q_0(n+1) = \overline{Q_0} \cdot \overline{Q_0} + \overline{Q_0} \cdot Q_0$$
$$= \overline{Q_0}$$

D



$$Q_0(n+1) = \overline{Q_0} \cdot \overline{Q_0} + \overline{1} \cdot Q_0$$
$$= \overline{Q_0}$$

Question MSQ (A, C, D)

T-FF
↳ $Q(n+1) = T \oplus Q(n)$

for this $A \odot B = T$

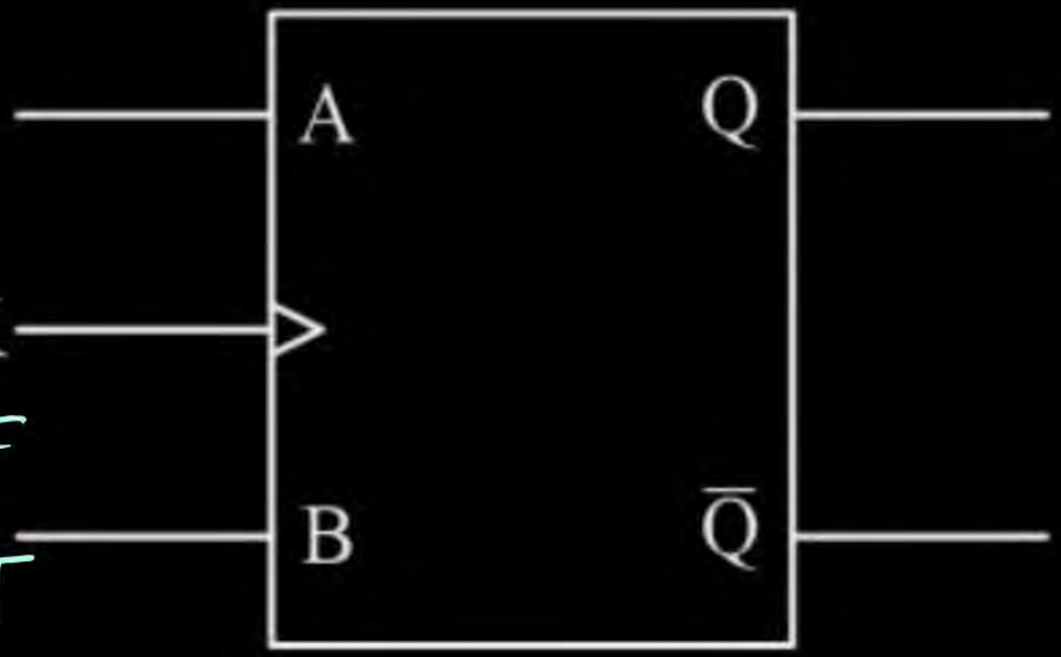
A FF is given as:

Characteristic equation of above FF is

$Q(n+1) = A \odot B \oplus Q(n)$

To convert this FF into T-FF,

A & B will be:



A. $A = T, B = 1$ $A \odot B = T \odot 1 = T$

B. $A = \overline{T}, B = 1$ $\overline{T} \odot 1 = T \oplus 1 = \overline{T}$

C. $A = \overline{T}, B = 0$ $\overline{T} \odot 0 = T \oplus 0 = T$

D. $A = 0, B = \overline{T}$ $0 \odot \overline{T} = T$

Question MCQ



To convert J-K FF into D-FF, J-K input will be

A $J = \bar{D}, K = D$

B $J = K = D$

C $J = D, K = \bar{D}$

D $J = K = \bar{D}$

JK to D $\rightarrow$ $J = D$, $K = \bar{D}$

JK to T $\rightarrow$ $J = T$, $K = T$

$Q(n+1) = J\bar{Q} + \bar{K}Q = D\bar{Q} + DQ = D[\bar{Q} + Q]$

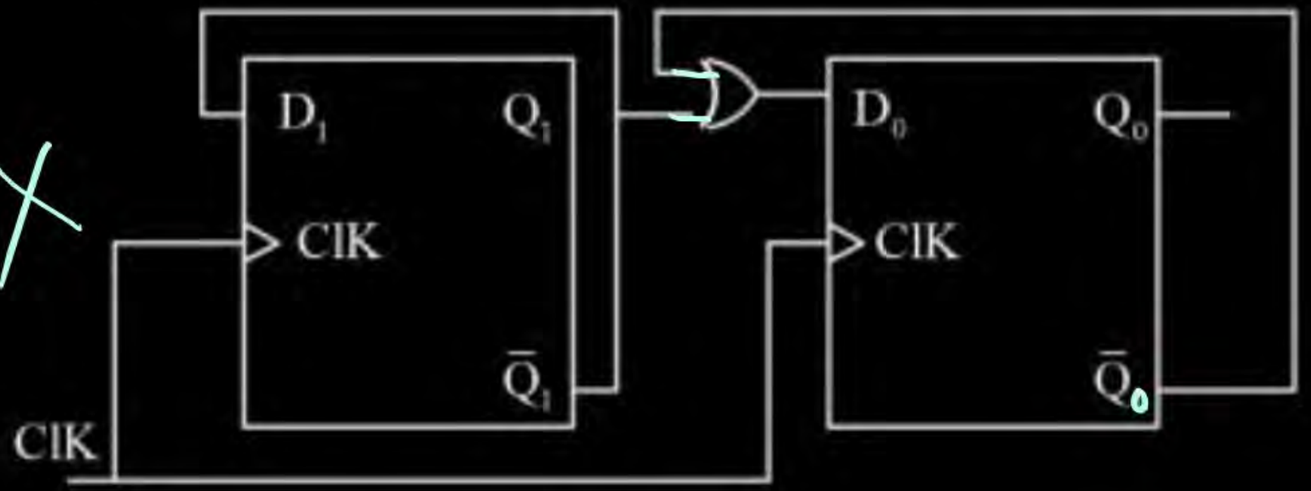
$= D \cdot 1 = D$

Question MCQ

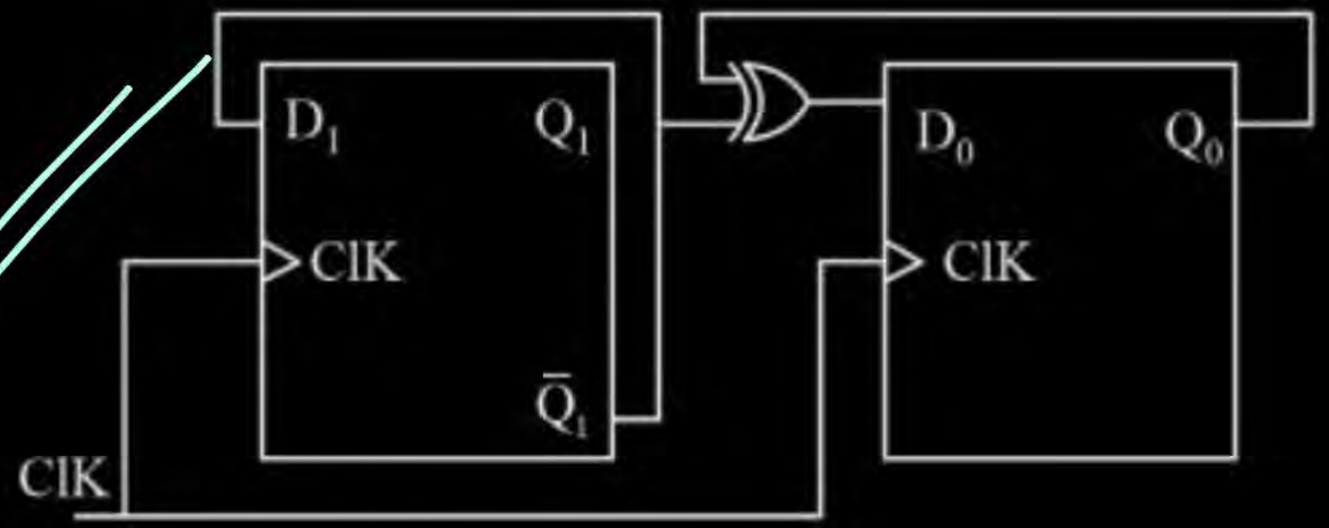
★★★

To design a counter with counting sequence 0 – 3 – 1 – 2 – 0 the circuit will be

A ✗

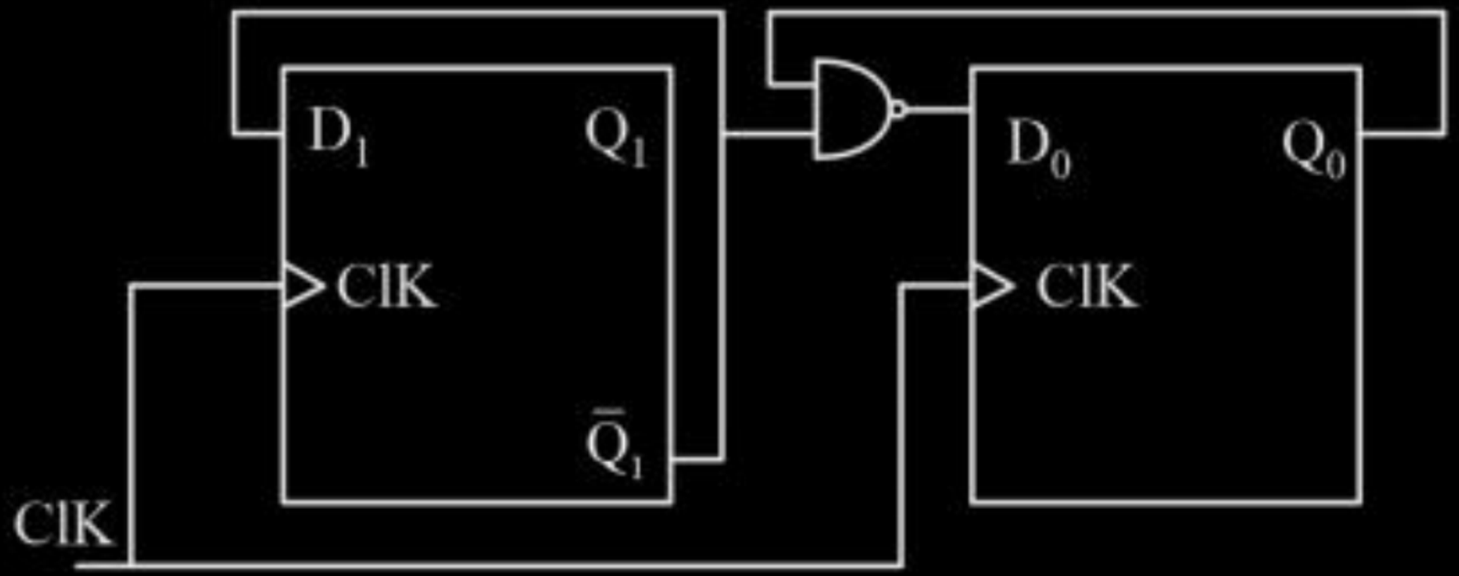


B



Q_1	Q_0	$Q_0(n+1)$
0	0	$= D_0$
1	1	$= \bar{Q}_0 + \bar{Q}_1$
0	1	$= \overline{Q_0 \cdot Q_1}$
1	0	$Q_0(n+1)$
0	0	$= Q_0 \oplus \bar{Q}_1$
		$= Q_0 \odot Q_1$

C

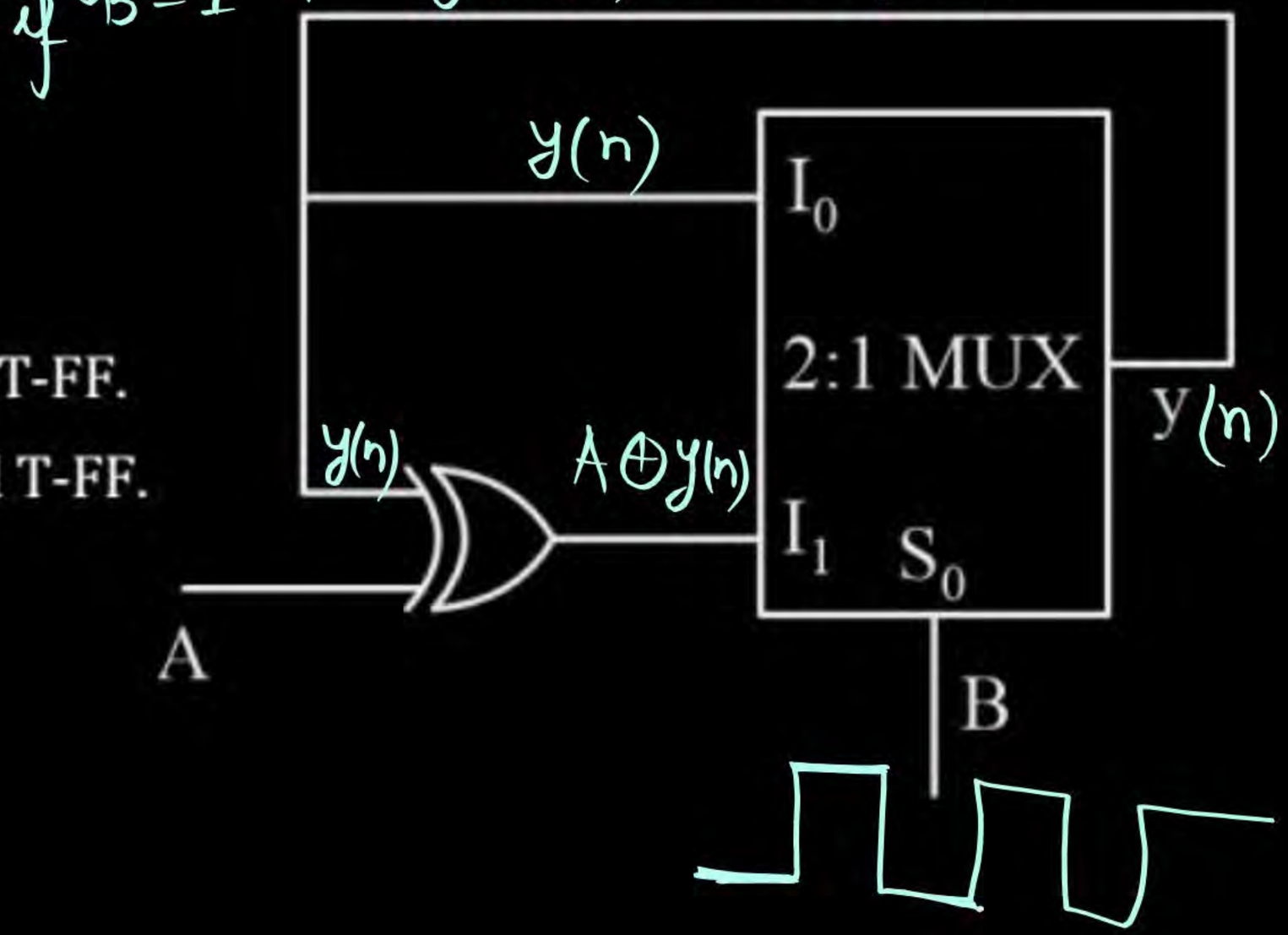


D None of these

Question MCQ

A circuit is as given below:

The circuit can work as



A. Positive level triggered T-FF.

B. Negative level triggered T-FF.

C. Level triggered D-FF.

D. Edge triggered T-FF

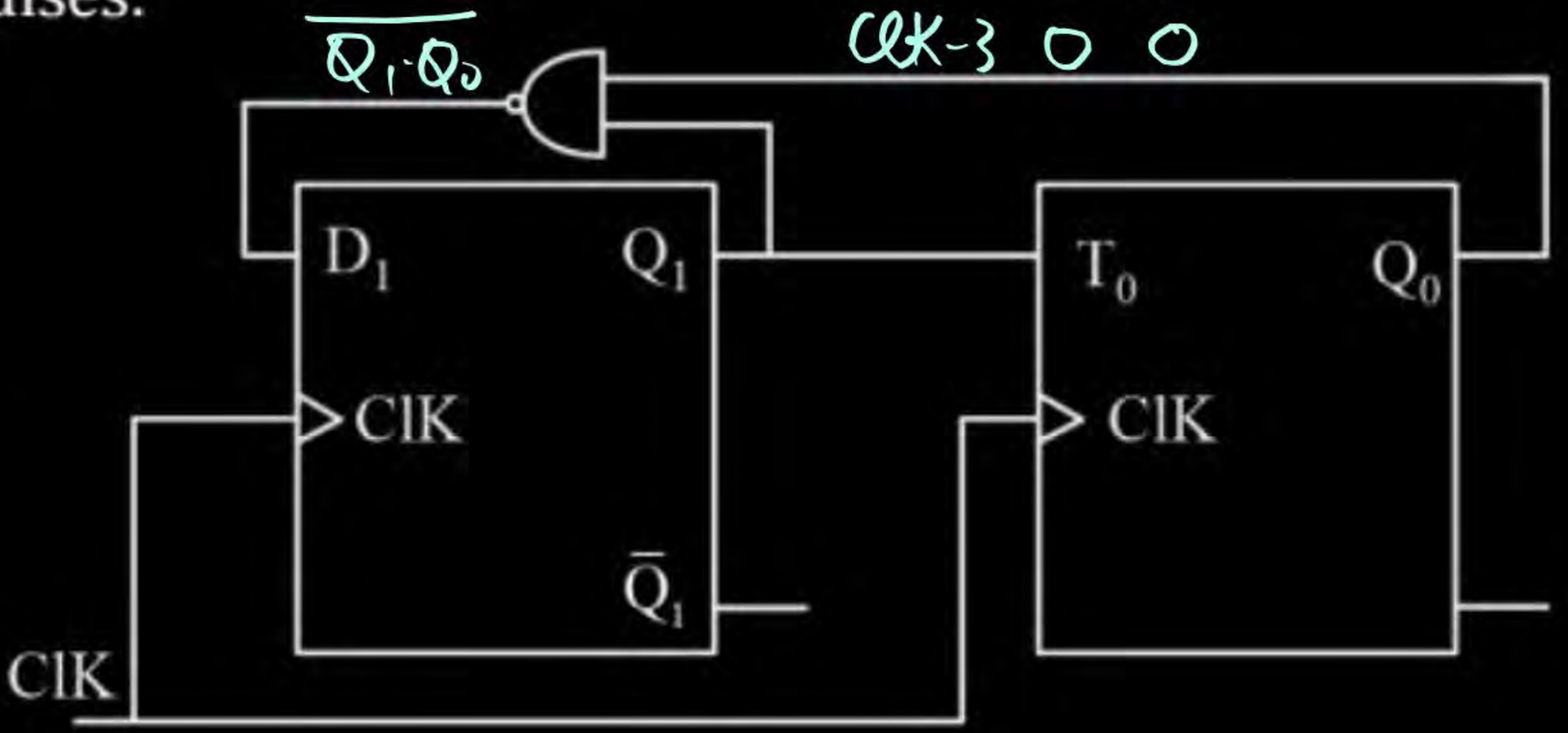
Question MCQ

$Q_0(n+1) = T_0 \oplus Q_0 =$
$= Q_1 \oplus Q_0$
$Q_1(n+1) = D_1 = \overline{Q_1 \cdot Q_0}$

	Q_0	Q_1
	0	0
CLK-1	0	1
CLK-2	1	1
CLK-3	0	0

A circuit is as given below:

Initially FFs are at rest $Q_0Q_1 = (00)_2$, then sequence that will be generated after application of clock pulses:



A. 00 – 10 – 11 – 00

B. 00 – 01 – 10 – 11 – 00

C. 00 – 01 – 11 – 00

D. 00 – 01 – 11 – 10 – 00

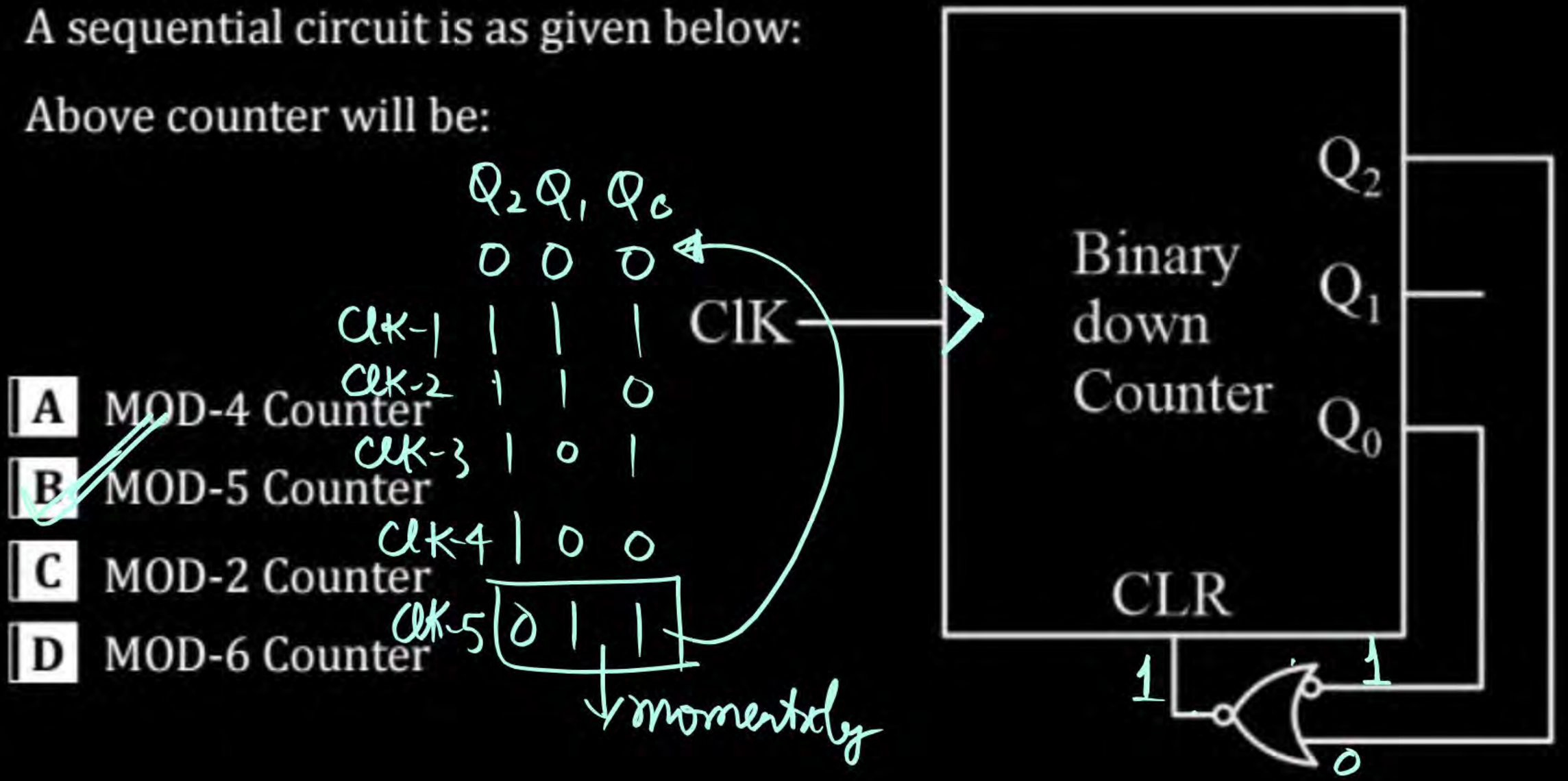
Question MCQ

★★☆

$Q_2 = 0$ & $Q_0 = 1$ → CLR pin activated

A sequential circuit is as given below:

Above counter will be:



A MOD-4 Counter

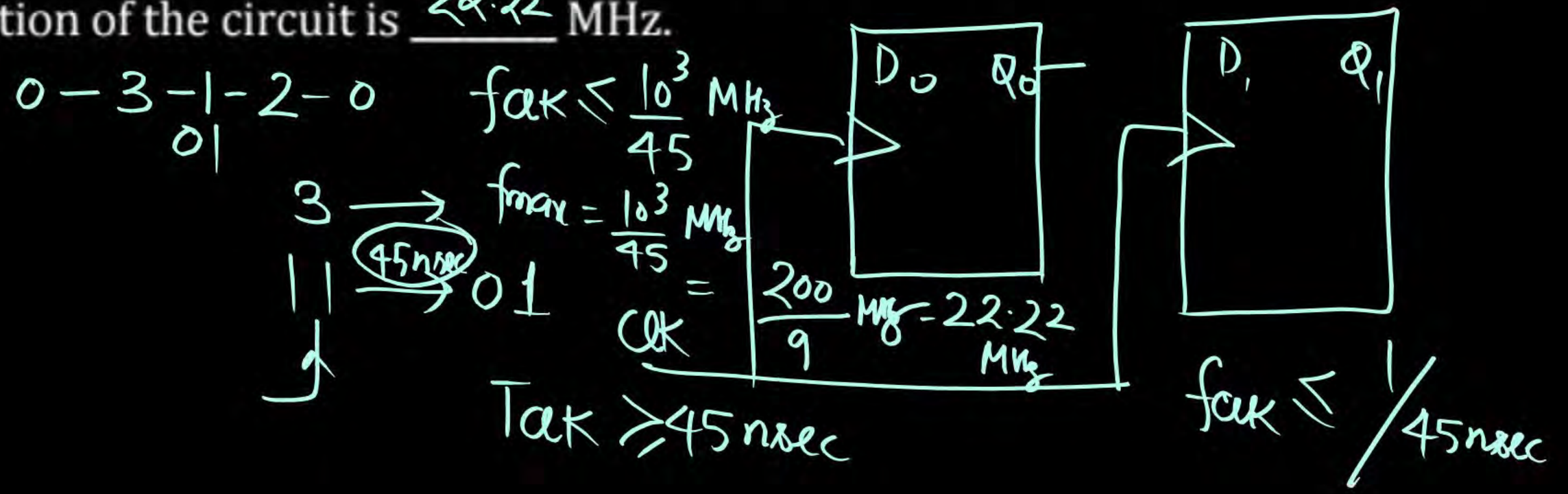
B MOD-5 Counter

C MOD-2 Counter

D MOD-6 Counter

Question **NAT**

A synchronous counter is designed using D-FFs that generate sequence 0 – 3 – 1 – 2 – 0. FFs used have delay of t_d = 45 nsec and logic gates used has zero delay, then maximum clock frequency that can be used to have proper operation of the circuit is 22.22 MHz.



Question MCQ

$Q_1(n+1) = J_1\overline{Q_1} + \overline{K_1}Q_1$

$= \overline{Q_0}\cdot\overline{Q_1} + Q_0Q_1 = Q_1 \odot Q_0$

$Q_0(n+1) = J_0\overline{Q_0} + \overline{K_0}Q_0 = \overline{Q_1}\,\overline{Q_0} + 0 = \overline{Q_1 + Q_0}$

A sequential circuit is as given below:

Initially counter is at $Q_1Q_0 = (11)_2$ then after applying clock next state will be

$Q_1\ Q_0$

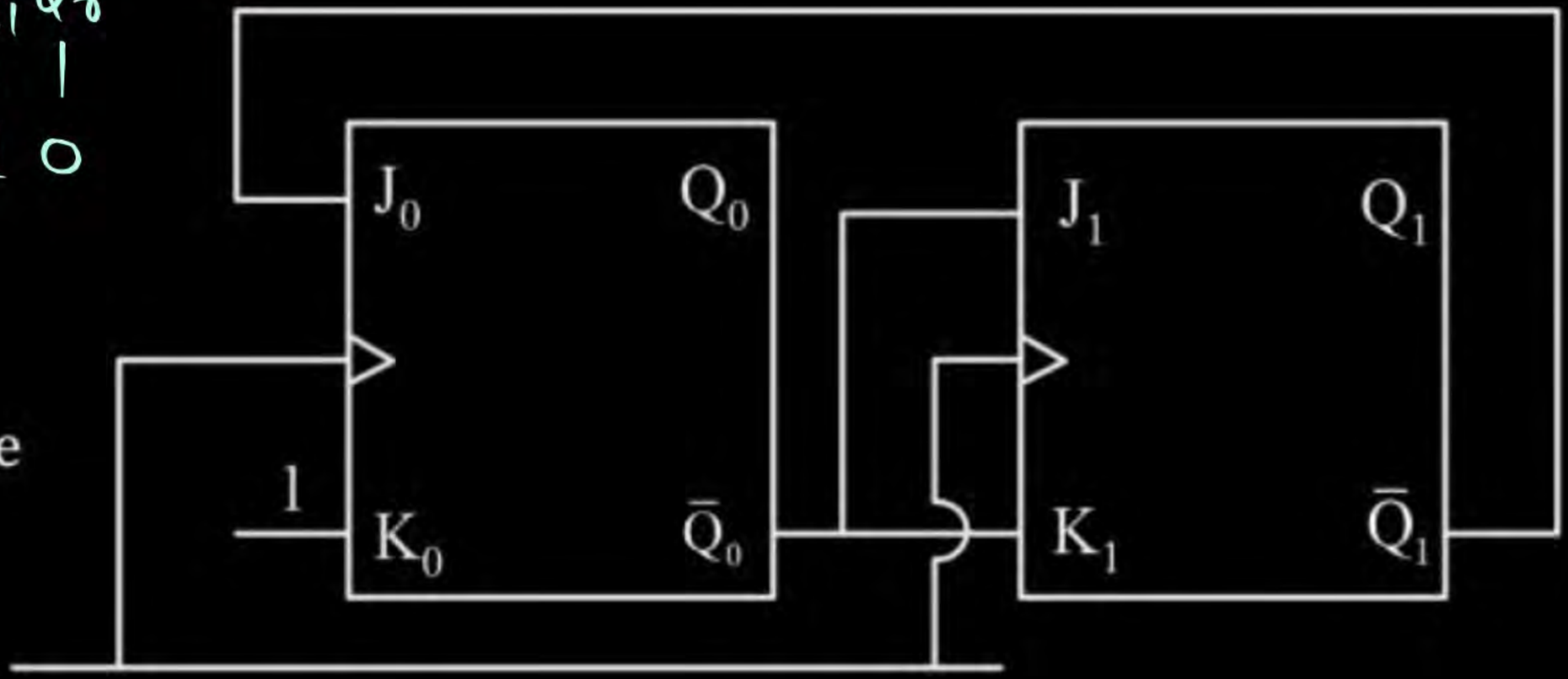
Clk-1 → 1 0

A $(01)_2$

B $(10)_2$

C $(00)_2$

D None of these



Question MCQ

We have a T-FF, to convert it into D-FF, input T must be

$T \Rightarrow \quad Q(n+1) = T \oplus Q(n)$

DFF $\rightarrow Q(n+1) = D$

A $D \odot Q$ ✗

B $D \oplus Q$ ✓

C D ✗

D $\overline{D}$ ✗

$T = D \oplus Q(n) \Rightarrow Q(n+1) = \underline{D \oplus Q(n)} \oplus Q(n)$

$Q(n+1) = D \oplus 0 = D$

Question **MCQ**

We want to design a counter with counting sequence 0 – 1 – 2 – 3 – 0, using
Circuit-A → Asynchronous counter $\longrightarrow FF_s = 2 \rightarrow T_d = 50\,nsec$
Circuit-B → Synchronous counter $\rightarrow FF_s = 2$
$(T_{clk})_A \geq (50\,nsec)$
FFs used in circuit-A and circuit-B are of same logic family and logic gates used has 0 delay while FFs have delay of 25 nsec. Then which of the following is true?

$(T_d)_B = 25\,nsec$
$(T_{clk})_B \geq 25\,nsec$
$(f_{clk})_A \leq 1/50\,nsec$
$(f_{clk})_{A\,max} = 1/50\,nsec$

A ✗ Maximum clock frequency is same in both the circuits.

B ✓ Maximum clock frequency is double in circuit-B compare to circuit-A.

C ✗ Maximum clock frequency is half in circuit-B compare to circuit-A

D None of these

$(f_{clk})_B \leq 1/25\,nsec$ $(f_{clk})_{B\,max} = 1/25\,nsec = 2 \cdot 1/50\,nsec$

Question NAT

★★☆

A counter has to be designed to generate a sequence $0 - 2 - 2^2 - 2^3 - 2^4 - 2^5 - 2^6$, to design the counter appropriate number of FFs needed is 7.

0 – 2 – 4 – 8 – 16 – 32 – (64) MOD no = 7

$2^6 = 64$

6 → FF → MOD no = 64

0 — (63)

2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	0	0	0	0	0	0

$= (64)_{10}$

$2^7 = \underline{128}$

Thank you

GW

Soldiers !

